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PREDICTIVE CONTROL OF SWITCHED AFFINE SYSTEMS SUBJECTED TO DWELL TIME CONSTRAINTS

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PREDICTIVE CONTROL OF SWITCHED AFFINE SYSTEMS SUBJECTED TO DWELL TIME CONSTRAINTS

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ITA

dedico esse trabalho Deus, e a minha familia e amigos

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"I am and always will be the optimist.
The hoper of far-flung hopes, and the dreamer of improbable dreams."
— The Doctor

Resumo

Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o sistema à trajetória cíclica alvo.

Abstract

Switched actuators are difficult to handle because of some characteristic constraints of their nature. But they are efficient in numerous engineering applications, being applied, for example, in attitude control systems for space vehicles. Thus, knowing how to use this type of actuator is important and necessary. The main problem to be solved in this thesis is the design of a controller that guarantees the robustness of a system with switched actuators subjected to time constraints that limit the response window of their commands. In this approach, a limit cycle target trajectory search is performed and then the error of the states compared to the target trajectory is calculated. Then a predictive control technique is applied to correct the error and get the system to the cyclic target trajectory

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List of Abbreviations and Acronyms

DC	Direct Current
DLQR	Discrete-time Linear Quadratic Regulator
MILP	Mixed-Integer Linear Programming
MPC	Model Predictive Control
SQP	Sequential Quadratic Programming
MAS	Maximum Admissible Set

List of Symbols

A	dynamics matrix
A_i	i-th dynamics matrix A
b	dynamics matrix
b_i	i-th dynamics matrix b
$e[j]$	state error at j-th cycle
J	cost function
k	current discrete time
N_p	prediction horizon
Ω^{∞}	optimal sequence of modes
Q, R	design weights
$\sigma(t)$	operation mode over time
σ_i	i-th operation mode
s^{∞}	maximum number of modes
T_p	cycle period
$T_{p,max}$	maximum cycle period
$\Delta t_{i,min}$	minimum duration time between switch i
τ^{∞}	optimal time sequence
t	time
t_i	i-th perturbed switching time
$\bar{t}_i$	i-th switching time
u	control signal
x	state vector
$\bar{x}$	reference state vector
$\bar{x}^{\infty}$	average reference value
$\hat{x}$	predicted state vector
$x(t)$	state vector continous
$x[k]$	k-th state vector
x_0	initial state vector

Contents

1 Introduction

1.1 Motivation

The dynamics of switched affine systems can be described by a state equation of the form $\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}, t \in \mathbb{R}_+$, where $x(t) \in \mathbb{R}^n$ is the state vector and $\sigma(t) \in \{1, 2, ..., M\}$ is the switching state associated with the matrices $A_{\sigma(t)} \in \mathbb{R}^{n\times n}$ and the vectors $b_{\sigma(t)} \in \mathbb{R}^n$ at each time t.

There is extensive research in the synthesis of controllers for switched affine systems. However, the design procedures are typically concerned with driving the state $x(t)$ to the vicinity of a given target, without considering the subsequent evolution of the state in that region. This issue is especially relevant when the switching state $\sigma(t)$ is subjected to dwell-time constraints. In that case, rather than driving the state to an equilibrium point, the design should be aimed at achieving a suitable limit cycle.

This problem has been addressed, for instance, by Benmiloud *et al.* (2019) who uses a hybrid Poincaré map approach to stabilize a switched affine system into the desired limit cycle and a DC-DC converter was employed as an illustrative example. Egidio *et al.* (2020b) proposed the control design of a state-dependent switching function for discrete-time switched affine systems to stabilize a limit cycle determined according to criteria of interest related to the steady-state behavior. The control design applies a switched rule to orchestrate the state trajectories based on a time-varying convex Lyapunov function that drives the system from the initial condition toward the desired limit cycle The proposed method uses a voltage regulation of a simulated DC-DC multicellular converter as an illustrative example. This work was extended to continuous time switched-affine systems by Egidio *et al.* (2020a)

Some studies form the basis for the present work:

Vieira (2013) solved the problem using predictive control to account for the dwell-time constraints of switching affine systems. Here the controller drove the system to a target state using MILP (Mixed-Integer Linear Programming) and the cyclic trajectory was a consequence of the states as the system approached the target.

Patino *et al.* (2010) defined the target trajectory using nonlinear programming and used this target trajectory in the predictive control in terms of errors.

Marcolino (2018) proposed a solution with a controller law that drives the switched affine system with dwell-time constraints to a target cyclic trajectory. In this work, the problem is addressed in two stages: first, the problem is interpreted as a convex optimization where the author can find a targeted cyclic trajectory that minimizes a cost function associated with the system performance; then a second part that uses the target trajectory as input and computes perturbation at the switching instants of the reference control signal using predictive control.

1.2 Contributions of the present work

The present work proposes an extension of the solution from Marcolino *et al.* (2021) to a general case of switched affine systems with changing dynamics at switching times and applies the dwell-time constraints using predictive control.

The work proposed by Marcolino *et al.* (2021) has an approach with three basic elements: the determination of a target cyclic trajectory, a linearization of the cycle-to-cycle dynamics of the system around the target trajectory that applies the control in the switching time perturbation, the use of the linearized model with predictive formulation to impose the dwell time constraints. This formulation of the solution also has the convenience of solving the problem with quadratic programming convex optimization.

1.3 Outline of the remaining chapters

The remainder of this text is organized as follows. Chapter 2 describes the problem of finding a target periodic trajectory for the given switched affine system. Chapter 3 proposes a predictive control law that drives the system to the target cyclic trajectory. Finally, chapter 4 brings some remarks and considerations of the work.

2 Determination of Target Periodic Trajectories

Consider a switching affine system of the form

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)},$$

the goal is to drive a system defined by an affine dynamic equation to an optimal periodic trajectory, so the first step is to generate this trajectory. This trajectory is submitted to restrictions imposed by the dynamic of the system and also switching time restrictions.

2.1 Optimization

Following the approach from Patino *et al.* (2010) we must establish a trajectory that yields the optimal time sequence τ^∞, the modes sequence Ω^∞, and its length $s^\infty(1 < s^\infty < s_{max})$. To reduce the oscillations around a desired operation point, we choose an integral quadratic criterion centered on the average reference value $\bar{x}^\infty$:

$$J(\tau^\infty, \Omega^\infty, s^\infty, x_0) = \min_{\tau,\Omega,s} \int_0^{T_p} \Big[x(t) - \bar{x}^\infty\Big]^T Q \Big[x(t) - \bar{x}^\infty\Big] dt \tag{2.1}$$

where $\tau^\infty = \big[t_1, t_2, \cdots, t_{s^\infty}\big]$, $\Omega^\infty = \big[\sigma_1, \sigma_2, \cdots, \sigma_{s^\infty}\big]$, $Q = Q^T > 0$ and T_p is the period of the cycle.

Since the trajectory is cyclic we can impose the constraint:

$$x(0) = x(T_p) \tag{2.2}$$

2.1.1 System Dynamics and Sequence

We are considering that the system dynamics will change after each switch command, this means that each switching interval will have its dynamics represented by the matrices

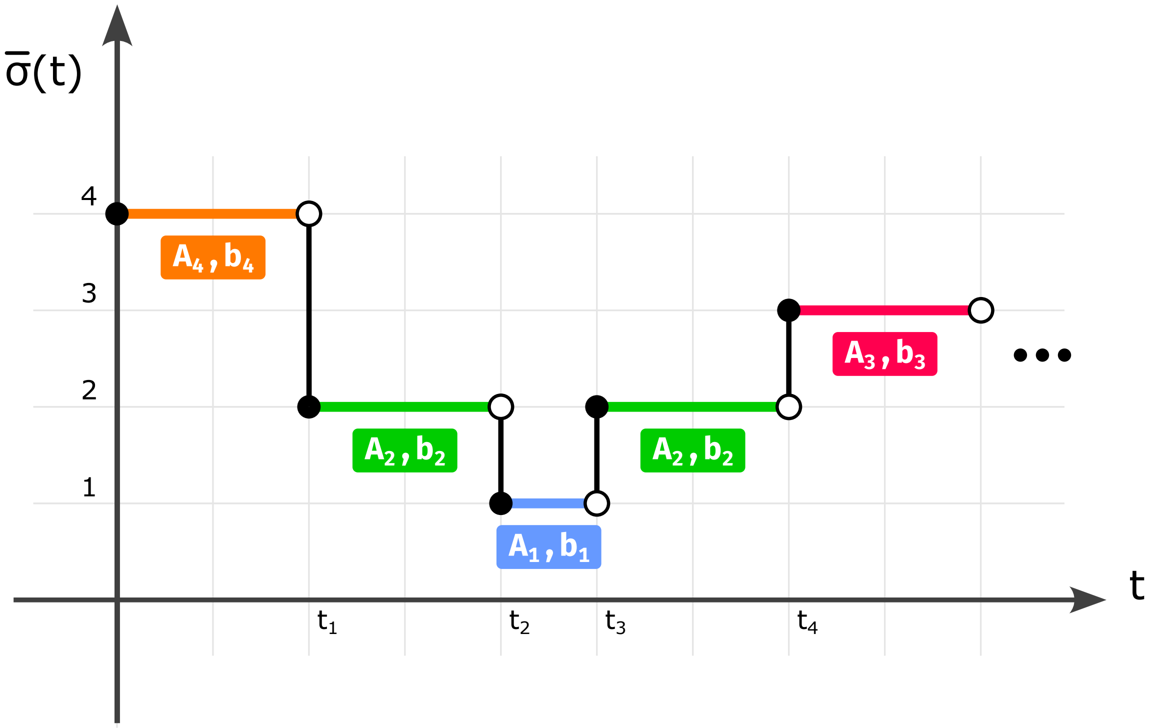

FIGURE 2.1 – Change in dynamics over trajectory time

$\{A_i, b_i\}$. The same system dynamic can be used again in different intervals. This is illustrated in figure 2.1, where a sequence of dynamics is applied in the sequence of modes $[4, 2, 1, 2, 3]$ to generate the cyclic trajectory.

2.1.2 Constraints Definition

Once the list and sequence have been established, we need to apply the constraints, these constraints dictate the cycle's maximum duration

$$T_p \leq T_{p,max} \tag{2.3}$$

as well as the minimum time permitted between switches in modes in the form of

$$(t_i - t_{i-1}) \geq \Delta t_{i,min} \ , \quad i = 1, 2, \ldots, N \tag{2.4}$$

where T_p is the cycle period, $T_{p,max}$ is the maximum cycle period, $\Delta t_{i,min}$ is the minimum duration time permitted between switch i.

2.2 Calculating x_0

We need to compute the initial condition for the system for each evaluation of the cost function so that we have a periodic trajectory, meaning that the last states of the trajectory are the same as the initial condition.

We can have different dynamics for each switching instant, so the system behaves as

$$\begin{aligned} t \in \left[0, t_1\right] &: \dot{x} = A_{\sigma(0)} x + b_{\sigma(0)} \\ t \in \left[t_1, t_2\right] &: \dot{x} = A_{\sigma(t_1)} x + b_{\sigma(t_1)} \\ &\vdots \\ t \in \left[t_{N-1}, t_N\right] &: \dot{x} = A_{\sigma(t_{N-1})} x + b_{\sigma(t_{N-1})} \end{aligned} \tag{2.5}$$

We can rewrite the solution for the first region as

$$\begin{aligned} x(t_1) &= e^{A_{\sigma(0)} t_1} x(0) + \int_0^{t_1} e^{A_{\sigma(0)}(t_1 - \tau)} b_{\sigma(0)} d\tau \\ x(t_1) &= F_0 x(0) + g_0 \end{aligned} \tag{2.6}$$

then the behavior of the other regions of the system is

$$\begin{aligned} x(t_1) &= F_0 x(0) + g_0 \\ x(t_2) &= F_1 x(t_1) + g_1 \\ &\vdots \\ x(t_N) &= F_{N-1} x(t_{N-1}) + g_{N-1} \end{aligned} \tag{2.7}$$

with

$$F_i = e^{A_{\sigma(t_i)}(t_{i+1} - t_i)} \tag{2.8}$$

$$g_i = \int_{t_i}^{t_{i+1}} e^{A_{\sigma(t_i)}(t_{i+1} - \tau)} b_{\sigma(t_i)} d\tau = \int_0^{t_{i+1} - t_i} e^{A_{\sigma(t_i)} \tau} b_{\sigma(t_i)} d\tau \tag{2.9}$$

From (2.7) we can establish the relation between $x(t_N)$ and $x(0)$

$$\begin{aligned} x(t_N) = {} & F_{N-1} F_{N-2} \ldots F_0 x(0) + \\ & F_{N-1} F_{N-2} \ldots F_1 g_0 + F_{N-1} F_{N-2} \ldots F_2 g_1 + \ldots + F_{N-1} g_{N-2} + g_{N-1} \\ x(t_N) = {} & F_{N-1} F_{N-2} \ldots F_0 x(0) + c \end{aligned} \tag{2.10}$$

We know that $x(t_N) = x(0)$, so the solution for $x(0)$ is

$$\begin{aligned}
x(0) &= F_{N-1}F_{N-2}\dots F_0 x(0) + c \\
\Big(I - F_{N-1}F_{N-2}\dots F_0\Big)x(0) &= c \\
x(0) &= \Big(I - F_{N-1}F_{N-2}\dots F_0\Big)^{-1} c
\end{aligned} \tag{2.11}$$

where

$$c = F_{N-1}F_{N-2}\dots F_1 g_0 + F_{N-1}F_{N-2}\dots F_2 g_1 + \dots + F_{N-1}g_{N-2} + g_{N-1} \tag{2.12}$$

assuming that the inverse indicated in (2.11) exists.

2.2.1 Optimization with FMINCON

For optimization purposes, the function FMINCON is employed. This function works in conjunction with a cost function. Its primary goal is to evaluate and discern the most cost-effective result that respects the imposed constraints in (2.3) and (2.4).

The optimization algorithm solves the problem

$$\min_x f(x) \text{ such that } \begin{cases} c(x) \leq 0 \\ ceq(x) = 0 \\ A \cdot x \leq B \\ Aeq \cdot x = beq \\ lb \leq x \leq ub \end{cases}$$

FMINCON is a method of optimization in MATLAB's libraries that implements the method SQP (Sequential Quadratic Programming). The basic SQP algorithm is described in Chapter 18 of Nocedal e Wright (2006).

The cost function $f(x)$ is designed to accept increments of switching times constrained by $lb \leq x$ to evaluate the trajectory with the dwell-time constraints in place.

2.3 Application Examples

2.3.1 Buck-Boost Converter

We are using an application example extracted from Patino *et al.* (2010), a Buck-Boost converter in Continuous Conduction Mode (CCM) with the topology shown in Figure 2.2.

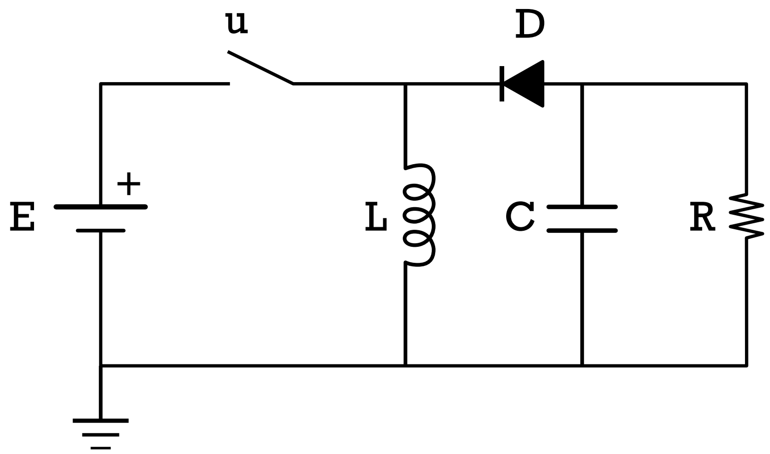

FIGURE 2.2 – Buck-Boost Converter

The state variables are the capacitor voltage v_c and the inductor current i_L. Its dynamics are given by the equation:

$$\dot{x} = \begin{cases} A_1 x + b_1, \text{ for u} = 1 \text{ (closed switch - Mode 1)} \\ A_2 x + b_2, \text{ for u} = 0 \text{ (open switch - Mode 2)} \end{cases} \tag{2.13}$$

where

$$\begin{gathered} A_1 = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{RC} \end{bmatrix} \quad A_2 = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \\ b_1 = \begin{bmatrix} \frac{E}{L} & 0 \end{bmatrix}^T \qquad b_2 = \begin{bmatrix} 0 & 0 \end{bmatrix}^T \end{gathered} \tag{2.14}$$

where (in normalized values) $R = L = C = E = 1$.

This example has two dynamics:

$$\left\{ \left[A_1, b_1\right], \left[A_2, b_2\right] \right\} \tag{2.15}$$

where

$$A_1 = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, b_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \tag{2.16}$$

$$A_2 = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}, b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{2.17}$$

The criteria for the optimization is to compute the trajectory to minimize equation (2.1) with $\bar{x}^\infty = \left[2\text{ V}, -1\text{ A}\right]^T$ with the weight matrix $Q = diag\left[1, 1\right]$, the maximum cycle period of $T_{p,max} = 1\text{ s}$ and minimum dwell time of $t_{min} = 0.25\text{ s}$.

The optimization generates the switching time duration and this relates to τ^∞ as

$$\tau^\infty = \left[t_0, t_1, t_2\right] = \left[t_0, t_0 + dx[1], t_1 + dx[2]\right] \tag{2.18}$$

where dx is the array of switching time duration and it's constrained by a lower bound that has to be greater or equal to t_{min}

$$dx = \begin{bmatrix} dx_1 \\ dx_2 \end{bmatrix} \geq \begin{bmatrix} t_{min} \\ t_{min} \end{bmatrix} = \begin{bmatrix} 0.25 \\ 0.25 \end{bmatrix} \tag{2.19}$$

and constrained by a maximum cycle time $T_{p_{max}}$

$$\left[1, \cdots, 1\right] \cdot \begin{bmatrix} dx_1 \\ dx_2 \end{bmatrix} \leq T_{p_{max}} = 1.0 \tag{2.20}$$

The solution provided by the FMINCON optimization yields

$$\begin{gathered} \tau^\infty = \left[0, 0.2513, 0.5914\right], \\ \Omega^\infty = \left[1, 2\right]. \end{gathered} \tag{2.21}$$

The outcome under this condition is displayed in Figure 2.3, showcasing the cyclic trajectory around the average value $[2, -1]^T$. The corresponding switch command signal u can be seen in Figure 2.4.

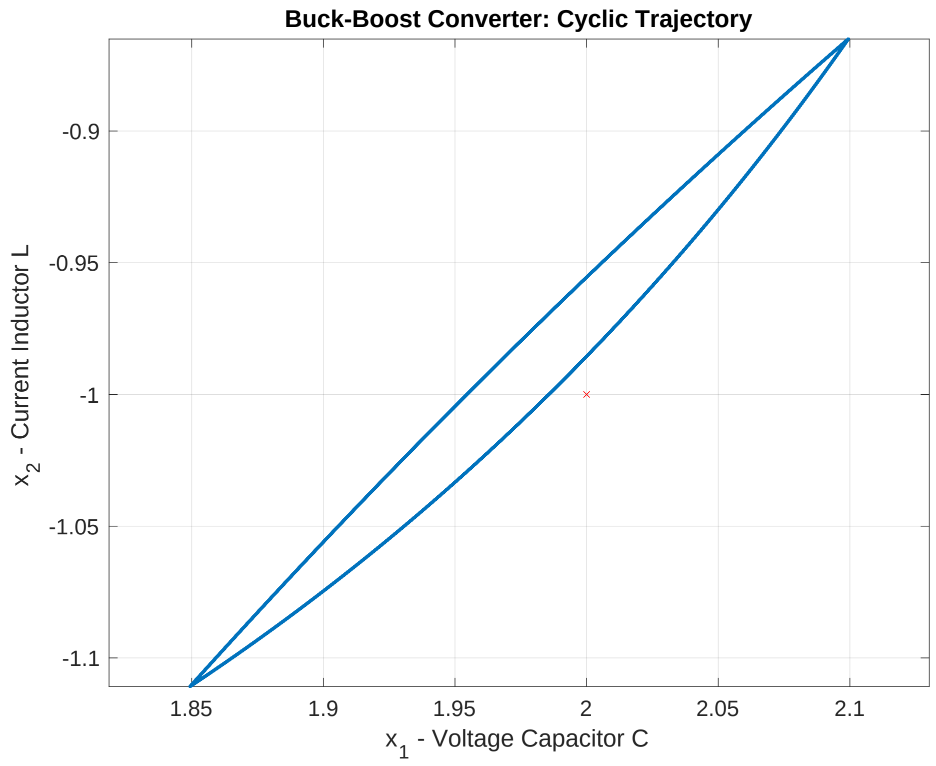

FIGURE 2.3 – Buck-Boost Converter: State trajectory

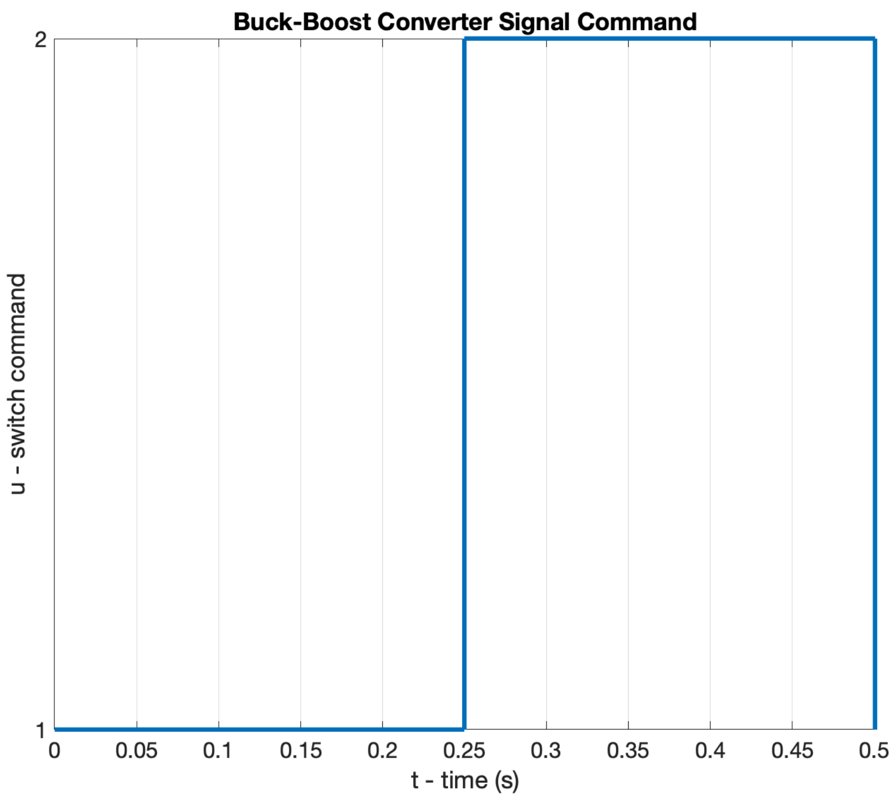

FIGURE 2.4 – Buck-Boost Converter: Switch command

2.3.2 Multilevel Converter

The second application example is also extracted from Patino *et al.* (2010), a Multilevel converter with topology is shown in Figure 2.5.

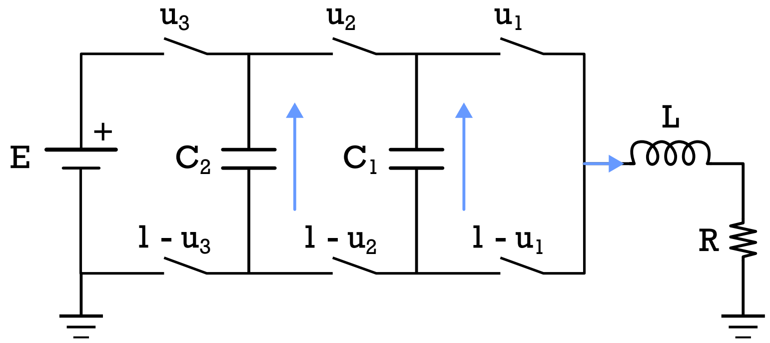

FIGURE 2.5 – Four-level Three-cell Converter

The state vector is the voltage on the capacitors $x(t) = [v_{c_1}(t), v_{c_2}(t), i_L(t)]$, where v_{c_1}, v_{c_2} are the voltages in each capacitor and i_L is the load current.

The dynamic of this system can be described as

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & \frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\frac{R}{L}x_3(t) \end{bmatrix} \tag{2.22}$$

rearranging the terms we have

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 0 & \frac{u_2-u_1}{C1} \\ 0 & 0 & \frac{u_3-u_2}{C2} \\ \frac{u_1-u_2}{L} & \frac{u_2-u_3}{L} & \frac{-R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\frac{R}{L}u_3 \end{bmatrix} \tag{2.23}$$

Each combination of the switches determines a different behavior:

$$
\begin{aligned}
\mathbf{u_1} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad A_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{-R}{L} \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
\mathbf{u_2} &= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_1} \\ 0 & \frac{-1}{L} & \frac{-R}{L} \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} \\
\mathbf{u_3} &= \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & \frac{-1}{C_2} \\ \frac{-1}{L} & \frac{1}{L} & \frac{-R}{L} \end{bmatrix}, \quad b_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
\mathbf{u_4} &= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad A_4 = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & 0 \\ \frac{-1}{L} & 0 & \frac{-R}{L} \end{bmatrix}, \quad b_4 = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} \\
\mathbf{u_5} &= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad A_5 = \begin{bmatrix} 0 & 0 & \frac{-1}{C_1} \\ 0 & 0 & 0 \\ \frac{1}{L} & 0 & \frac{-R}{L} \end{bmatrix}, \quad b_5 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
\mathbf{u_6} &= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad A_6 = \begin{bmatrix} 0 & 0 & \frac{-1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ \frac{1}{L} & \frac{-1}{L} & \frac{-R}{L} \end{bmatrix}, \quad b_6 = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} \\
\mathbf{u_7} &= \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad A_7 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{-1}{C_1} \\ 0 & \frac{1}{L} & \frac{-R}{L} \end{bmatrix}, \quad b_7 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
\mathbf{u_8} &= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad A_8 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{-R}{L} \end{bmatrix}, \quad b_8 = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix}
\end{aligned}
\tag{2.24}
$$

With the detailed dynamic for each switching position, we can define the list of elements for the multilevel converter as $\Big\{A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8\Big\}$ and $\Big\{b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8\Big\}$.

The criteria for the optimization is to compute the trajectory around the average value $\Big[\frac{1}{3}E, \frac{2}{3}E, 1\Big]$ with $E = 30$ V. The maximum cycle period is $T_{p,max} = 0.4$ ms and minimum dweel time of $t_{min} = 0.022$ ms.

The parameters for the simulation are $C_1 = 40\ \mu$F, $C_2 = 40\ \mu$F, $L = 10$ mH, $R = 10\ \Omega$ with the weight matrix $Q = diag\big[10, 5, 20000\big]$ end implements the mode sequence from Patino *et al.* (2010)

$$\Omega^\infty = \Big\{1, 2, 4, 8, 3, 1, 5, 8, 5\Big\}$$

The constraints of the switching time duration for this example are

$$dx = \begin{bmatrix} dx_1 \\ dx_2 \\ \vdots \\ dx_9 \end{bmatrix} \geq \begin{bmatrix} t_{min} \\ t_{min} \\ \vdots \\ t_{min} \end{bmatrix} = \begin{bmatrix} 0.022 \\ 0.022 \\ \vdots \\ 0.022 \end{bmatrix} \text{ ms} \tag{2.25}$$

for the minimum switching time duration t_{min}, and

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix} \cdot \begin{bmatrix} dx_1 \\ dx_2 \\ \vdots \\ dx_9 \end{bmatrix} \leq T_{p_{max}} = 0.4 \text{ ms} \tag{2.26}$$

for the maximum cycle duration $T_{p_{max}}$.

The solution provided by the FMINCON optimization yields

$$\tau^{\infty} = \left\{0, 0.064, 0.088, 0.112, 0.134, 0.157, 0.206, 0.230, 0.253, 0.276\right\} \text{ ms}$$

with the initial condition of the cyclic trajectory

$$x(0) = \left[9.514 \text{ V}, 19.821 \text{ V}, 1.031 \text{ A}\right]^T$$

The outcome under this condition is displayed in sequence, Figure 2.6 displays the change of the states x_1, x_2 and x_3 over time, Figure 2.7 displays the cyclic trajectory of the states and Figure 2.8 shows the resulting input signal for the cyclic trajectory.

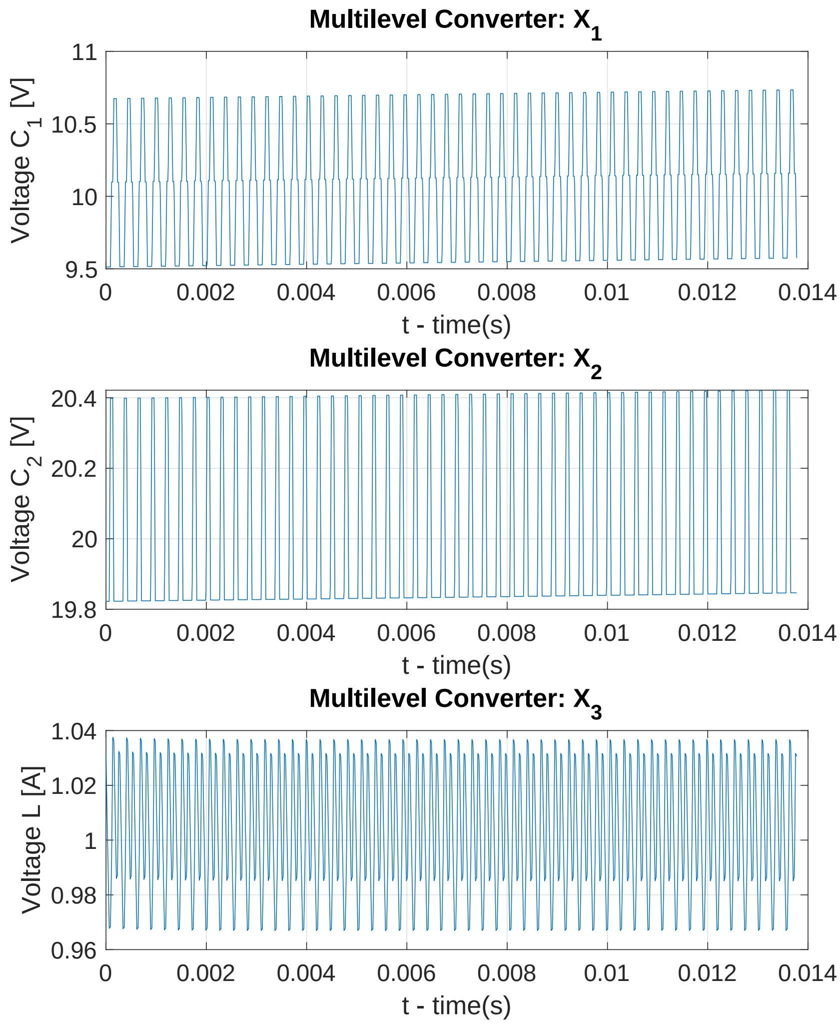

FIGURE 2.6 – Multilevel Converter: States over Time

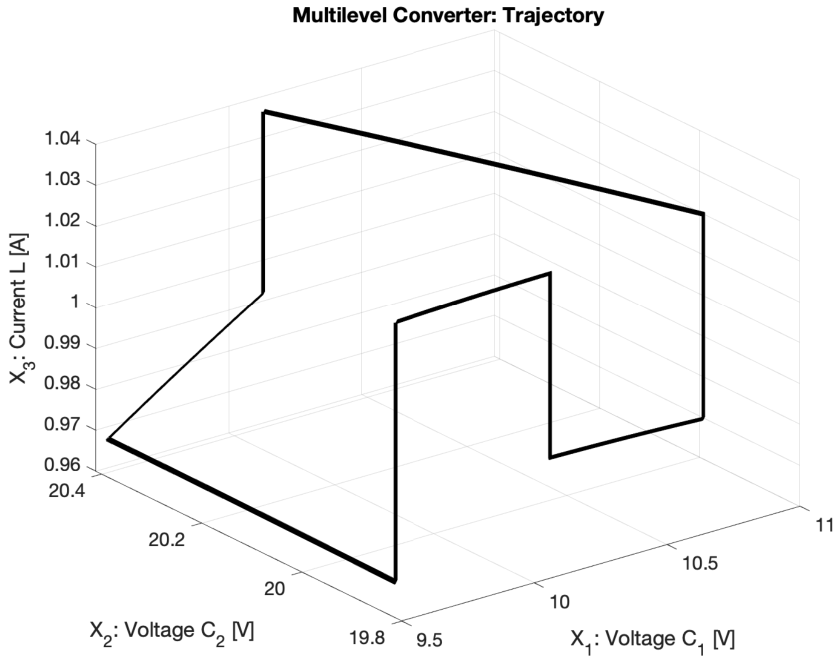

FIGURE 2.7 – Multilevel Converter: State Trajectory

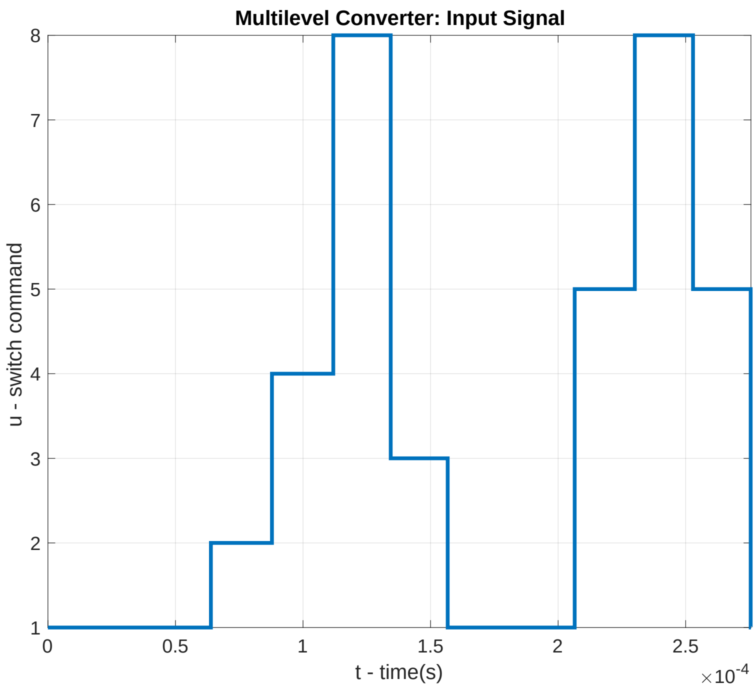

FIGURE 2.8 – Multilevel Converter: Input Signal

3 Predictive control: Preliminaries and notation

Consider a system with discrete-time dynamics described by

$$x[k+1] = Ax[k] + Bu[k] \tag{3.1}$$

where $x[k] \in \mathbb{R}^n$ and $u[k] \in \mathbb{R}^p$ are the state and control vectors at the kth time instant.

The goal is to drive the state $x[k]$ to the origin from a given initial condition while respecting state and control constraints in the form

$$x[k] \in \mathcal{X} \tag{3.2}$$

$$u[k] \in \mathcal{U} \tag{3.3}$$

where $\mathcal{U}$ and $\mathcal{X}$ are polyhedral sets containing the origin in their interior. It is assumed that $\mathcal{U}$ and $\mathcal{X}$ are described in half-space representation as

$$\mathcal{U} = \{u \in \mathbb{R}^p : S_u u \leq b_u\} \tag{3.4}$$

and

$$\mathcal{X} = \{x \in \mathbb{R}^n : S_x x \leq b_x\} \tag{3.5}$$

with S_u, b_u and S_x, b_x of suitable dimensions.

3.1 Cost Function

At each time k, the control action $u[k]$ will be determined by solving an an optimization problem with the cost given by

$$J = \left\|\hat{x}[k+N|k]\right\|_{P_f}^2 + \sum_{i=0}^{N-1}\left[\left\|\hat{x}[k+i|k]\right\|_Q^2 + \left\|\hat{u}[k+i|k]\right\|_R^2\right] \tag{3.6}$$

where $\hat{x}[k+i|k]$ are predicted values of the state and control vectors, which are related through the following equations:

$$\hat{x}[k|k] = x[k] \tag{3.7}$$

$$\hat{x}[k+i+1|k] = A\hat{x}[k+i|k] + B \tag{3.8}$$

$$\hat{x}[k+i|k], \quad i = 0, 1, \cdots, N-1 \tag{3.9}$$

The expression for J consists of a terminal cost term and a trajectory cost term.

With P_f, Q, R being symmetric and positive-definite, with P_f chosen so that the terminal cost $\left\|x\right\|_{P_f}^2$ term corresponds to the value function of an infinite horizon Discrete-Time Linear Quadratic Regulator (DLQR) designed width weights Q, R.

By using the notation

$$\mathbf{\hat{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \in \mathbb{R}^{pN}, \mathbf{\hat{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \in \mathbb{R}^{nN} \tag{3.10}$$

the cost function can be written in the form

$$J(\mathbf{\hat{x}}[k], \mathbf{\hat{u}}[k]) = \left\|x[k]\right\|_Q^2 + \left\|\mathbf{\hat{x}}[k]\right\|_{\mathbf{Q}}^2 + \left\|\mathbf{\hat{u}}[k]\right\|_{\mathbf{R}}^2 \tag{3.11}$$

where

$$\mathbf{Q} = \begin{bmatrix} Q & 0 & \cdots & 0 \\ 0 & Q & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & P_f \end{bmatrix} = \begin{bmatrix} diag_{N-1}(Q) & 0 \\ 0 & P_f \end{bmatrix} \in \mathbb{R}^{nN\times nN} \tag{3.12}$$

and

$$\mathbf{R} = \begin{bmatrix} R & 0 & \cdots & 0 \\ 0 & R & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R \end{bmatrix} \in \mathbb{R}^{pN \times pN} \tag{3.13}$$

3.2 Constraints

In view of (3.2) and (3.3), the constraints on the optimization variables ca be expressed as

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \in \mathcal{U} \times \mathcal{U} \times \cdots \times \mathcal{U} = \mathcal{U}^N \tag{3.14}$$

$$\hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \in \mathcal{X} \times \mathcal{X} \times \cdots \times \mathcal{X}_f = \mathcal{X}^{N-1} \times \mathcal{X}_f \tag{3.15}$$

where

$$\mathcal{X}_f = \left\{ x \in \mathbb{R}^n : S_f x \leq b_f \right\} \tag{3.16}$$

is the maximum invariant and admissible set (MAS) concerning the terminal state and control constraints under the infinite-horizon DLQR control law.

The constraint on $\hat{\mathbf{u}}[k]$ can be written as

$$\begin{bmatrix} S_u & 0 & \cdots & 0 \\ 0 & S_u & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_u \end{bmatrix} \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \leq \begin{bmatrix} b_u \\ b_u \\ \vdots \\ b_u \end{bmatrix} \tag{3.17}$$

or, in a shorter form

$$\mathbf{S_u} \hat{\mathbf{u}}[k] \leq \mathbf{b_u} \tag{3.18}$$

with

$$\mathbf{S_u} = diag_N(S_u), \mathbf{b_u} = col_N(b_u) \tag{3.19}$$

Similarly, in light of (3.5), the constraints on $\hat{\mathbf{x}}[k]$ can be written as

$$\begin{bmatrix} S_x & 0 & \cdots & 0 \\ 0 & S_x & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_x \end{bmatrix} \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \leq \begin{bmatrix} b_x \\ b_x \\ \vdots \\ b_x \end{bmatrix} \tag{3.20}$$

or, in short

$$\mathbf{S_x}\hat{\mathbf{x}}[k] \leq \mathbf{b_x} \tag{3.21}$$

with

$$\mathbf{S_x} = \begin{bmatrix} diag_{N-1}(S_x) & 0 \\ 0 & S_f \end{bmatrix}, \mathbf{b_x} = \begin{bmatrix} col_{N-1}(b_x) \\ b_f \end{bmatrix} \tag{3.22}$$

3.3 Prediction Equation

Let's express the relationship between the vectors

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix}, \hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \tag{3.23}$$

based on the prediction equation

$$\hat{x}[k+i+1|k] = A\hat{x}[k+i|k] + B\hat{u}[k+i|k], i = 0, 1, \dots, N \tag{3.24}$$

and initial condition $\hat{x}[k|k] = x[k]$.

From (3.24), it follows that

$$\hat{x}[k+i|k] = A^i\hat{x}[k|k] + \sum_{j=0}^{i-1} A^{i-1-j}B\hat{u}[k+j|k] \tag{3.25}$$

which can be written for $i = 1, 2, ..., N$ as

$$\hat{x}[k+1|k] = Ax[k] + B\hat{u}[k|k] \tag{3.26}$$

$$\hat{x}[k+2|k] = A^2x[k] + AB\hat{u}[k|k] + B\hat{u}[k+1|k] \tag{3.27}$$

$$\vdots \tag{3.28}$$

$$\hat{x}[k+N|k] = A^Nx[k] + A^{N-1}B\hat{u}[k|k] + \cdots + AB\hat{u}[k+N-2|k] + B\hat{u}[k+N-1|k] \tag{3.29}$$

or

$$\begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix} x(0) + \begin{bmatrix} B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \cdots & B \end{bmatrix} \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \tag{3.30}$$

This can be written in a shorter form as

$$\hat{\mathbf{x}}[k] = \mathbf{A}x[k] + \mathbf{B}\hat{\mathbf{u}}[k] \tag{3.31}$$

with

$$\mathbf{A} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix}, \mathbf{B} = \begin{bmatrix} B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \cdots & B \end{bmatrix} \tag{3.32}$$

The vectors $\hat{\mathbf{x}}[k]$ and $\hat{\mathbf{u}}[k]$ are linked by (3.31), so the constraint (3.21) can be rewritten as

$$\mathbf{S_x B u} \leq \mathbf{b_x} - \mathbf{S_x A}x(0) \tag{3.33}$$

Combined with $\mathbf{S_u u}[k] \leq \mathbf{b_u}$, it can be expressed in the form

$$\mathbf{S u}[k] \leq \mathbf{b} \tag{3.34}$$

with

$$\mathbf{S} = \begin{bmatrix} \mathbf{S_x B} \\ \mathbf{S_u} \end{bmatrix}, \mathbf{b} = \begin{bmatrix} \mathbf{b_x} - \mathbf{S_x A}\hat{x}[k|k] \\ \mathbf{b_u} \end{bmatrix} \tag{3.35}$$

3.4 Optimization Problem

We can rewrite the cost function as

$$J = \left\| x[k|k] \right\|_P^2 + \overbrace{\left\| \hat{\mathbf{x}}[k] \right\|_{\mathbf{Q}}^2}^{\square} + \left\| \hat{\mathbf{u}}[k] \right\|_{\mathbf{R}}^2 \tag{3.36}$$

$$\begin{aligned}
\square &= \left\|\hat{\mathbf{x}}[k]\right\|_{\mathbf{Q}}^2 && (3.37)\\
&= \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}}[k]\right]^T \mathbf{Q}\left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}}[k]\right] && (3.38)\\
&= \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}}[k]\right]^T \mathbf{Q}\left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}}[k]\right] && (3.39)\\
&= \left\|\mathbf{A}x[k|k]\right\|_{\mathbf{Q}}^2 + \left[\mathbf{A}x[k|k]\right]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}}[k] \\
&\quad + \hat{\mathbf{u}}[k]^T\mathbf{B}^T\mathbf{Q}\mathbf{A}x[k|k] + \hat{\mathbf{u}}[k]^T\mathbf{B}^T\mathbf{Q}\mathbf{B}\hat{\mathbf{u}}[k] && (3.40)
\end{aligned}$$

replacing $\square$ back in the cost equation and reordering the elements

$$\begin{aligned}
J(\hat{\mathbf{x}}, \hat{\mathbf{u}}[k]) = &\left\|x[k|k]\right\|_P^2 + \left\|\mathbf{A}x[k|k]\right\|_{\mathbf{Q}}^2 && (3.41)\\
&+ \left[\mathbf{A}x[k|k]\right]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}}[k] + \hat{\mathbf{u}}[k]^T\mathbf{B}^T\mathbf{Q}\mathbf{A}x[k|k] && (3.42)\\
&+ \hat{\mathbf{u}}[k]^T\mathbf{R}\hat{\mathbf{u}}[k] + \hat{\mathbf{u}}[k]^T\mathbf{B}^T\mathbf{Q}\mathbf{B}\hat{\mathbf{u}}[k] && (3.43)
\end{aligned}$$

The constant part will be denoted by c_0

Recalling that the matrix $\mathbf{Q}$ is symmetric, we have:

$$\left[\mathbf{A}x[k|k]\right]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}}[k] = \left\{\hat{\mathbf{u}}[k]^T\mathbf{B}^T\mathbf{Q}\mathbf{A}x[k|k]\right\}^T \tag{3.44}$$

As these terms are scalars, it can be concluded that they are equal to each other, therefore:

$$\begin{aligned}
&\left[\mathbf{A}x[k|k]\right]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}}[k] + \hat{\mathbf{u}}[k]^T\mathbf{B}^T\mathbf{Q}\mathbf{A}x(0) \\
&\qquad = 2\left[\mathbf{A}x[k|k]\right]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}}[k]
\end{aligned} \tag{3.45}$$

Finally, it can be written

$$J(\hat{\mathbf{u}}[k]) = \hat{\mathbf{u}}[k]^T\left(\mathbf{R} + \mathbf{B}^T\mathbf{Q}\mathbf{B}\right)\hat{\mathbf{u}}[k] + 2\left[\mathbf{A}x[k|k]\right]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}}[k] + c_0 \tag{3.46}$$

with

$$c_0 = \left\|x(0)\right\|_P^2 + \left\|\mathbf{A}x(0)\right\|_{\mathbf{Q}}^2 \tag{3.47}$$

$$c_0 = \left\|x[k|k]\right\|_P^2 + \left\|\mathbf{A}x[k|k]\right\|_{\mathbf{Q}}^2$$

Note that we will now use the notation $J(\hat{\mathbf{u}}[k])$ instead of $J(\hat{\mathbf{x}}[k], \hat{\mathbf{u}}[k])$, since $\hat{\mathbf{x}}[k]$ has been eliminated from the cost expression.

To finalize the restructuring of the optimization problem in terms of $\hat{\mathbf{u}}[k]$, the relation between $\hat{\mathbf{x}}[k]$ and $\hat{\mathbf{u}}[k]$ in the restrictions must be considered.

The optimization problem can be expressed as

$$\min_{\hat{\mathbf{x}}[k]\in\mathbb{R}^{nN},\hat{\mathbf{u}}[k]\in\mathbb{R}^{pN}} J(\hat{\mathbf{x}}[k],\hat{\mathbf{u}}[k]) \tag{3.48}$$

where

$$J(\hat{\mathbf{x}}[k],\hat{\mathbf{u}}[k]) = \left\|x[k|k]\right\|_P^2 + \left\|\hat{\mathbf{x}}[k]\right\|_{\mathbf{Q}}^2 + \left\|\hat{\mathbf{u}}[k]\right\|_{\mathbf{R}}^2 \tag{3.49}$$

with

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \in \mathbb{R}^{pN}, \hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \in \mathbb{R}^{nN}, \tag{3.50}$$

subject to

$$\mathbf{S_x}\hat{\mathbf{x}}[k] \leq \mathbf{b_x} \tag{3.51}$$

$$\mathbf{S_u}\hat{\mathbf{u}}[k] \leq \mathbf{b_u} \tag{3.52}$$

Thus, the cost can be rewritten as

$$\begin{gathered} J(\hat{\mathbf{u}}[k]) = \hat{\mathbf{u}}[k]^T(\mathbf{R} + \mathbf{B}^T\mathbf{Q}\mathbf{B})\hat{\mathbf{u}}[k] \\ 2\Big[\mathbf{A}\hat{x}[k|k]\Big]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}}[k] + c_0 \end{gathered}$$

with

$$c_0 = \left\|\hat{x}[k|k]\right\|_P^2 + \left\|\mathbf{A}\hat{x}[k|k]\right\|_{\mathbf{Q}}^2$$

and the constraints can be expressed as

$$\mathbf{S}\hat{\mathbf{u}}[k] \leq \mathbf{b} \tag{3.53}$$

Having obtained the solution $\hat{\mathbf{u}}^*[k|k]$ for the problem $\mathbb{P}\Big(\hat{x}[k|k]\Big)$, the control to be applied to the plant is given by

$$\hat{u}[k] = \hat{u}^*[k|k]$$

where $\hat{u}^*[k|k]$ corresponds to the first p elements of $\hat{\mathbf{u}}^*[k|k]$.

3.5 Feasibility Region

We aim to determine the set $\mathcal{D}$ of points $x(0) \in \mathcal{X}$ for which there exists $\hat{\mathbf{u}}[\mathbf{k}] \in \mathbb{R}^{pN}$ satisfying the constraints

$$\begin{bmatrix} \mathbf{S_x B} \\ \mathbf{S_u} \end{bmatrix} \mathbf{u}[k] \leq \begin{bmatrix} \mathbf{b_x} - \mathbf{SxA}x(0) \\ \mathbf{b_u} \end{bmatrix} \tag{3.54}$$

These restrictions can be rewritten as

$$\begin{bmatrix} \mathbf{S_x B} & \mathbf{S_x A} \\ \mathbf{S_u} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{u}[k] \\ x(0) \end{bmatrix} \leq \begin{bmatrix} \mathbf{b_x} \\ \mathbf{b_u} \end{bmatrix} \tag{3.55}$$

adding a restriction $S_x x(0) \leq b_x$

$$\begin{bmatrix} \mathbf{S_x B} & \mathbf{S_x A} \\ \mathbf{S_u} & 0 \\ 0 & S_x \end{bmatrix} \begin{bmatrix} \mathbf{u}[k] \\ x(0) \end{bmatrix} \leq \begin{bmatrix} \mathbf{b_x} \\ \mathbf{b_u} \\ b_x \end{bmatrix} \tag{3.56}$$

This can be written in a shorter form

$$S_w w \leq b_w \tag{3.57}$$

with

$$w = \begin{bmatrix} \mathbf{u}[k] \\ x(0) \end{bmatrix}, S_w = \begin{bmatrix} \mathbf{S_x B} & \mathbf{S_x A} \\ \mathbf{S_u} & 0 \\ 0 & S_x \end{bmatrix}, b_w = \begin{bmatrix} \mathbf{b_x} \\ \mathbf{b_u} \\ b_x \end{bmatrix} \tag{3.58}$$

The constraint $S_w w \leq b_w$ define a polyhedron $\mathcal{P}_w$ in the space $\mathbb{R}^{(pN+n)}$ corresponding to the variables $\mathbf{u}[k]$, $x(0)$.

The set $\mathcal{D}$ is obtained by projection $\mathcal{P}_w$ onto the subspace of its last n components.

The projection can be performed using the *projection* function from the *MPT Toolbox.*

3.6 Augmented States

Our hypothesis is that the control input will remain constant for N_d consecutive cycles. To incorporate this into our system model, we will expand the prediction equation over these N_d steps and then redefine the terms for simplification.

Starting with the standard discrete-time linear prediction equation 3.1:

$$x[k+1] = Ax[k] + Bu[k]$$

We can iteratively expand this equation for several steps, assuming the control input u remains constant at $u[k]$ for the next $N_d - 1$ steps:

$$\begin{aligned}
x[k+1] &= Ax[k] + Bu[k] \\
x[k+2] &= A^2x[k] + ABu[k] + Bu[k] \\
x[k+3] &= A^3x[k] + A^2Bu[k] + ABu[k] + Bu[k] \\
&\vdots \\
x[k+N_d] &= A^{N_d}x[k] + \left(A^{N_{d-1}} + A^{N_{d-2}} + \cdots + A^0\right)Bu[k] \qquad (3.59)
\end{aligned}$$

To simplify the notation, let's define new matrices A_b and B_b as:

$$A_b = A^{N_d} \qquad (3.60)$$

$$B_b = \left(A^{N_{d-1}} + A^{N_{d-2}} + \cdots + A^0\right)B \qquad (3.61)$$

Furthermore, let's shift the time index by blocks of N_d cycles. This implies the following relationship between the original time index k and the new time index k':

$$\begin{aligned}
k' - 1 &= k - N_d \\
k' &= k \\
k' + 1 &= k + N_d \\
&\vdots
\end{aligned}$$

Considering this shift, a transition of one step in the new time index k' corresponds to a transition of N_d steps in the original time index k. Therefore, the state at time $k' + 1$ (which corresponds to $k + N_d$) can be expressed based on the state and control at time k' (which is the same as k).

Let's consider the final prediction equation:

$$x[k+N_d] = A_bx[k] + B_bu[k] \qquad (3.62)$$

Using the new time index k', this becomes:

$$x[k'+1] = A_bx[k'] + B_bu[k'] \qquad (3.63)$$

Now, defining the augmented state vector $X_a\left[k'\right]$ as:

$$X_a\left[k'\right] = \begin{bmatrix} x\left[k'\right] \\ z\left[k'-1\right] \end{bmatrix} \tag{3.64}$$

with

$$z\left[k'-1\right] = u\left[k'-1\right] \tag{3.65}$$

we get to the augmented state transition equation:

$$X_a\left[k'+1\right] = \begin{bmatrix} A_b & B_b \\ 0 & 0 \end{bmatrix} X_a\left[k'\right] + \begin{bmatrix} 0 \\ 1 \end{bmatrix} z\left[k'\right] \tag{3.66}$$

where $z\left[k'\right] = u\left[k'\right]$.

We can replace the augmented state transition matrix

$$A_a = \begin{bmatrix} A_b & B_b \\ 0 & 0 \end{bmatrix} \tag{3.67}$$

and the augmented input matrix

$$B_a = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{3.68}$$

resulting in

$$X_a\left[k'+1\right] = A_b X_a\left[k'\right] + B_a z\left[k'\right] \tag{3.69}$$

4 Predictive Control for Stabilization of Periodic Trajectories

4.1 Statement of the problem

Considering a plant with switched-affine dynamics described by:

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)} \tag{4.1}$$

We assume the existence of a periodic switching signal $\bar{\sigma}(t)$ associated with a periodic trajectory $\bar{x}(t)$, as obtained in Chapter 2. Our objective is to introduce perturbations in the switching instants such that the trajectory $x(t)$ converges to the reference trajectory $\bar{x}(t)$.

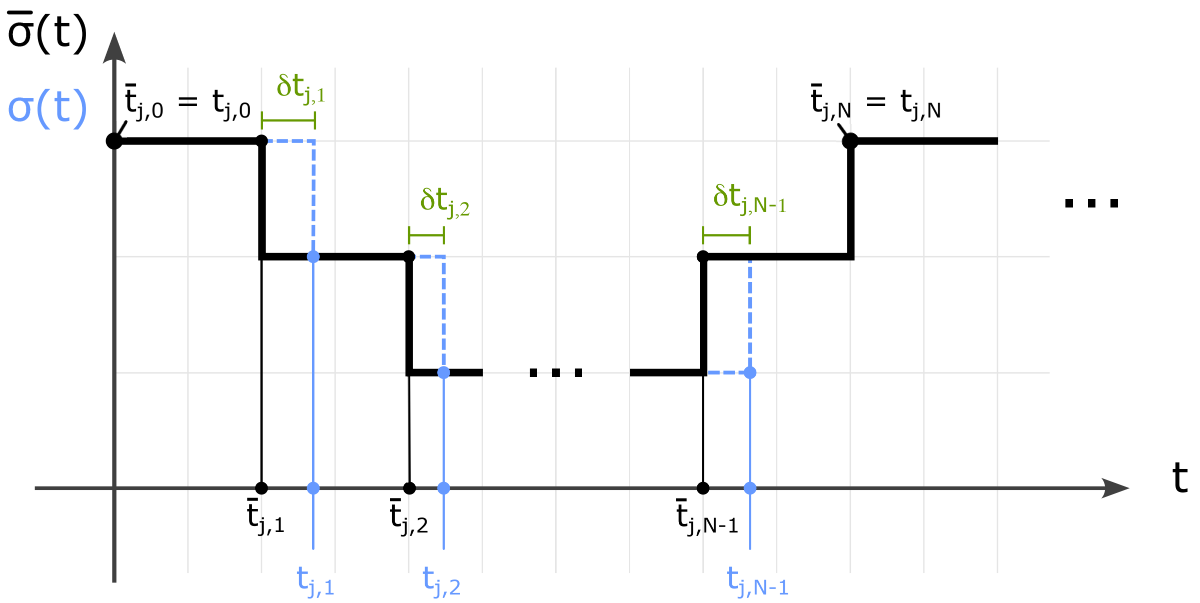

FIGURE 4.1 – Switching signals

Figure 4.1 displays the jth cycle of the switching signals $\bar{\sigma}(t)$ and $\sigma(t)$, which comprise N switching actions. Within this cycle, the i_{th} switching times of $\bar{\sigma}(t)$ and $\sigma(t)$ are denoted by $\bar{t}_{j,i}$ and $t_{j,i}$, respectively. The relation between $\bar{t}_{j,i}$ and $t_{j,i}$ is given by

$$t_{j,i} = \bar{t}_{j,i} + \delta t_{j,i} \tag{4.2}$$

In what follows, we derive a linear equation that relates the switching times and the resulting deviation of $x(t)$ with respect to $\hat{x}(t)$ at the end of one cycle. In the development of this linearization procedure, the j index will be dropped for brevity and simpler notation.

4.2 Linearization procedure

The solution for the state equation (4.1) at time t_1, starting from the initial condition $x(t_0)$, is of the form

$$x(t_1) = e^{A_1(t_1 - t_0)} x(t_0) + \int_{t_0}^{t_1} e^{A_1(t_1 - \tau)} B_1 d\tau \tag{4.3}$$

$$\bar{x}(\bar{t}_1) = e^{A_1(\bar{t}_1 - \bar{t}_0)} \bar{x}(\bar{t}_0) + \int_{\bar{t}_0}^{\bar{t}_1} e^{A_1(\bar{t}_1 - \tau)} B_1 d\tau \tag{4.4}$$

subtracting (4.4) from (4.3) we get

$$\begin{aligned} x(t_1) - \bar{x}(\bar{t}_1) = & \overbrace{e^{A_1(t_1 - t_0)} x(t_0) - e^{A_1(\bar{t}_1 - \bar{t}_0)} \bar{x}(\bar{t}_0)}^{\square} \\ & \underbrace{+ \int_{t_0}^{t_1} e^{A_1(t_1 - \tau)} B_1 d\tau - \int_{\bar{t}_0}^{\bar{t}_1} e^{A_1(\bar{t}_1 - \tau)} B_1 d\tau}_{\bigcirc} \end{aligned} \tag{4.5}$$

expanding (4.5) first in $\square$:

$$\begin{aligned} \square &= e^{A_1(t_1 - t_0)} x(t_0) - e^{A_1(\bar{t}_1 - \bar{t}_0)} \bar{x}(\bar{t}_0) \\ &= e^{A_1(\bar{t}_1 + \delta t_1 - t_0)} x(t_0) - e^{A_1(\bar{t}_1 - \bar{t}_0)} \bar{x}(\bar{t}_0) \\ &= e^{A_1(\bar{t}_1 - t_0)} e^{A_1 \delta t_1} x(t_0) - e^{A_1(\bar{t}_1 - \bar{t}_0)} \bar{x}(\bar{t}_0) \\ &= \bar{F}_1 e^{A_1 \delta t_1} x(t_0) - \bar{F}_1 \bar{x}(\bar{t}_0) \\ &= \bar{F}_1 (I + A_1 \delta t_1) x(t_0) - \bar{F}_1 \bar{x}(\bar{t}_0) \\ &= \bar{F}_1 [x(t_0) - \bar{x}(t_0)] + \bar{F}_1 A_1 \delta t_1 x(t_0) \end{aligned} \tag{4.6}$$

and then in $\bigcirc$:

$$
\begin{aligned}
\bigcirc &= \int_{t_0}^{t_1} e^{A_1(t_1-\tau)}B_1 d\tau - \int_{\bar{t}_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)}B_1 d\tau \\
&= \int_{t_0}^{\bar{t}_1+\delta t_1} e^{A_1(\bar{t}_1+\delta t_1-\tau)}B_1 d\tau - \int_{t_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)}B_1 d\tau \\
&= e^{A_1\delta t_1}\left[\int_{t_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)}B_1 d\tau + \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_1(\bar{t}_1-\tau)}B_1 d\tau\right] - \int_{t_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)}B_1 d\tau \\
&= e^{A_1\delta t_1}[\bar{G}_1 + B_1\delta t_1] - \bar{G}_1 \\
&= (I + A_1\delta t_1)[\bar{G}_1 + B_1\delta t_1] - \bar{G}_1 \\
&= \cancel{\bar{G}_1} + B_0\delta t_1 + A_1\bar{G}_1\delta t_1 + \cancel{A_1B_1\delta t_1\delta t_1} - \cancel{\bar{G}_1} \\
&= A_1\bar{G}_1\delta t_1 + B_1\delta t_1
\end{aligned}
\tag{4.7}
$$

the resulting equations (4.6) and (4.7) are replaced back in (4.5) to complete the difference equation

$$
\boxed{x(t_1) - \bar{x}(\bar{t}_1) = \bar{F}_1[x(t_0) - \bar{x}(t_0)] + [\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 + B_1]\delta t_1}
\tag{4.8}
$$

Repeating the steps for $x(t_2)$ we have

$$
x(t_2) = e^{A_2(t_2-t_1)}x(t_1) + \int_{t_1}^{t_2} e^{A_2(t_2-\tau)}B_2 d\tau
\tag{4.9}
$$

$$
\bar{x}(\bar{t}_2) = e^{A_2(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) + \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau
\tag{4.10}
$$

subtracting (4.10) from (4.9)

$$
\begin{aligned}
x(t_2) - \bar{x}(\bar{t}_2) = {}& \overbrace{e^{A_2(t_2-t_1)}x(t_1) - e^{A_2(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1)}^{\square} \\
& \underbrace{+ \int_{t_1}^{t_2} e^{A_2(t_2-\tau)}B_2 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau}_{\bigcirc}
\end{aligned}
\tag{4.11}
$$

expanding $\square$ in (4.11):

$$
\begin{aligned}
\square &= e^{A_2(\bar{t}_2+\delta t_2-\bar{t}_1-\delta t_1)}x(t_1) - e^{A_2(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) \\
&= e^{A_2(\bar{t}_2-\bar{t}_1)}e^{A_2(\delta t_2-\delta t_1)}x(t_1) - e^{A_2(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) \\
&= \bar{F}_2[I + A_2(\delta t_2 - \delta t_1)]x(t_1) - \bar{F}_2\bar{x}(\bar{t}_1) \\
&= \bar{F}_2[x(t_1) - \bar{x}(\bar{t}_1)] + \bar{F}_2 A_2 x(t_1)\delta t_2 - \bar{F}_2 A_2 x(t_1)\delta t_1
\end{aligned}
\tag{4.12}
$$

and expanding ◯:

$$\begin{aligned}
\bigcirc &= \int_{t_1}^{t_2} e^{A_2(t_2-\tau)}B_2 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau \\
&= \int_{\bar{t}_1+\delta t_1}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_2+\delta t_2-\tau)}B_2 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau \\
&= e^{A_2\delta t_2}\left[\int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau - \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau + \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau\right] \\
&\quad - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau \\
&= (I + A_2\delta t_2)\left[\bar{G}_2 - \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau + \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau\right] - \bar{G}_2 \\
&= (I + A_2\delta t_2)\left[\bar{G}_2 - e^{A_2(\bar{t}_2-\bar{t}_1)}B_2\delta t_1 + B_2\delta t_2\right] - \bar{G}_2 \\
&= (I + A_2\delta t_2)\left[\bar{G}_2 - \bar{F}_2 B_2\delta t_1 + B_2\delta t_2\right] - \bar{G}_2 \\
&= \cancel{\bar{G}_2} - \bar{F}_2 B_2\delta t_1 + B_2\delta t_2 + A_2\bar{G}_2\delta t_2 - \cancel{\bar{G}_2} \\
&= A_2\bar{G}_2\delta t_2 + B_2\delta t_2 - \bar{F}_2 B_2\delta t_1
\end{aligned} \tag{4.13}$$

and replacing (4.12) and (4.13) back into equation (4.11) we get to the difference equation for $x(t_2) - \bar{x}(\bar{t}_2)$

$$\begin{aligned}
x(t_2) - \bar{x}(\bar{t}_2) =& \bar{F}_2[x(t_1) - \bar{x}_1(\bar{t}_1)] \\
&- \bar{F}_2[A_2 x(t_1) + B_2]\delta t_1 \\
&+ [\bar{F}_2 A_2 \underbrace{x(t_1)}_{\triangle} + A_2\bar{G}_2 + B_2]\delta t_2
\end{aligned} \tag{4.14}$$

from (4.14), △ can be expanded by replacing the resulting equation of (4.8)

$$\triangle = x(t_1) = \bar{x}(\bar{t}_1) + \bar{F}_1[x(t_0) - \bar{x}(t_0)] + [\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 + B_1]\delta t_1 \tag{4.15}$$

note that when △ is replaced in eq. (4.14) it will produce a product with δt_1 and δt_2 like

$$x(t_1)\delta t_1 = \bar{x}(\bar{t}_1)\delta t_1 + \bar{F}_1\overbrace{[x(t_0) - \bar{x}(t_0)]\delta t_1}^{\square} + [\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 + B_1]\overbrace{\delta t_1\delta t_1}^{\bigcirc}$$

both □ and ◯ are products of differences that produces a smaller number and they will be neglected in the first order aproximations

$$x(t_1)\delta t_1 \doteq \bar{x}(\bar{t}_1)\delta t_1 \tag{4.16}$$

replacing (4.8) and (4.15) in (4.14) it follows that

$$\begin{aligned} x(t_2) - \bar{x}(\bar{t}_2) =& \bar{F}_2\{\bar{F}_1[x(t_0) - \bar{x}(t_0) + [\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 + B_1]\delta t_1]\} \\ & - \bar{F}_2[A_2\bar{x}(t_1) + B_2]\delta t_1 \\ & + [\bar{F}_2 A_2 \bar{x}(t_1) + A_2\bar{G}_2 + B_2]\delta t_2 \end{aligned}$$

$$\boxed{\begin{aligned} x(t_2) - \bar{x}(\bar{t}_2) =& \bar{F}_2\bar{F}_1[x(t_0) - \bar{x}(t_0)] \\ & + \bar{F}_2[\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 - A_2\bar{x}(\bar{t}_1) + B_1 - B_2]\delta t_1 \\ & + [\bar{F}_2 A_2\bar{x}(\bar{t}_1) + A_2\bar{G}_2 + B_2]\delta t_2 \end{aligned}} \tag{4.17}$$

Repeating the procedure for $x(t_3)$ we have

$$x(t_3) = e^{A_3(t_3-t_2)}x(t_2) + \int_{t_2}^{t_3} e^{A_3(t_3-\tau)}B_3 d\tau \tag{4.18}$$

$$\bar{x}(\bar{t}_3) = e^{A_3(\bar{t}_3-\bar{t}_2)}\bar{x}(\bar{t}_2) + \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)}B_3 d\tau \tag{4.19}$$

subtracting (4.18) from (4.19)

$$\begin{aligned} x(t_3)- > -\bar{x}(\bar{t}_3) = & \overbrace{e^{A_3(t_3-t_2)}x(t_2) - e^{A_3(\bar{t}_3-\bar{t}_2)}\bar{x}(\bar{t}_2)}^{\square} \\ & \underbrace{+\int_{t_2}^{t_3} e^{A_3(t_3-\tau)}B_3 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)}B_3 d\tau}_{\bigcirc} \end{aligned} \tag{4.20}$$

expanding $\square$ in (4.20):

$$\begin{aligned} \square &= e^{A_3(\bar{t}_3+\delta t_3-\bar{t}_2-\delta t_2)}x(t_2) - e^{A_3(\bar{t}_3-\bar{t}_2)}\bar{x}(\bar{t}_2) \\ &= e^{A_3(\bar{t}_3-\bar{t}_2)}e^{A_3(\delta t_3-\delta t_2)}x(t_2) - e^{A_3(\bar{t}_3-\bar{t}_2)}\bar{x}(\bar{t}_2) \\ &= \bar{F}_3[I + A_3(\delta t_3 - \delta t_2)]x(t_2) - \bar{F}_3\bar{x}(\bar{t}_2) \\ &= \bar{F}_3[x(t_2) - \bar{x}(\bar{t}_2)] + \bar{F}_3 A_3 x(t_2)\delta t_3 - \bar{F}_3 A_3 x(t_2)\delta t_2 \end{aligned} \tag{4.21}$$

and expanding $\bigcirc$:

$$
\begin{aligned}
\bigcirc &= \int_{t_2}^{t_3} e^{A_3(t_3-\tau)}B_3 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)}B_3 d\tau \\
&= \int_{\bar{t}_2+\delta t_2}^{\bar{t}_3+\delta t_3} e^{A_3(\bar{t}_3+\delta t_3-\tau)}B_3 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)}B_3 d\tau \\
&= e^{A_3\delta t_3}\left[\int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)}B_3 d\tau - \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_3(\bar{t}_3-\tau)}B_3 d\tau + \int_{\bar{t}_3}^{\bar{t}_3+\delta t_3} e^{A_3(\bar{t}_3-\tau)}B_3 d\tau\right] \\
&\quad - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)}B_3 d\tau \\
&= (I + A_3\delta t_3)\left[\bar{G}_3 - \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_3(\bar{t}_3-\tau)}B_3 d\tau + \int_{\bar{t}_3}^{\bar{t}_3+\delta t_3} e^{A_3(\bar{t}_3-\tau)}B_3 d\tau\right] - \bar{G}_3 \\
&= (I + A_3\delta t_3)\left[\bar{G}_3 - e^{A_3(\bar{t}_3-\bar{t}_2)}B_3\delta t_2 + B_3\delta t_3\right] - \bar{G}_3 \\
&= (I + A_3\delta t_3)\left[\bar{G}_3 - \bar{F}_3 B_3\delta t_2 + B_3\delta t_3\right] - \bar{G}_3 \\
&= \cancel{\bar{G}_3} - \bar{F}_3 B_3\delta t_2 + B_3\delta t_3 + A_3\bar{G}_3\delta t_3 - \cancel{\bar{G}_3} \\
&= A_3\bar{G}_3\delta t_3 + B_3\delta t_3 - \bar{F}_3 B_3\delta t_2
\end{aligned} \tag{4.22}
$$

and then replacing (4.21) in (4.22) back in eq. (4.20) we get to the difference equation for $x(t_3) - \bar{x}(\bar{t}_3)$

$$
\begin{aligned}
x(t_3) - \bar{x}(\bar{t}_3) =& \bar{F}_3[x(t_2) - \bar{x}_2(\bar{t}_2)] \\
& - \bar{F}_3[A_3 x(t_2) + B_3]\delta t_2 \\
& + [\bar{F}_3 A_3 x(t_2) + A_3\bar{G}_3 + B_3]\delta t_3
\end{aligned} \tag{4.23}
$$

from (4.17), $\triangle$ can be expanded the same way as in the past iteration and $x(t_2) - \bar{x}(\bar{t}_2)$ can be replaced in (4.23)

$$
\begin{aligned}
x(t_3) - \bar{x}(\bar{t}_3) =& \bar{F}_3\{\bar{F}_2\bar{F}_1[x(t_0) - \bar{x}(t_0)] \\
& + \bar{F}_2[\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 - A_2\bar{x}(\bar{t}_1) + B_1 - B_2]\delta t_1 \\
& + [\bar{F}_2 A_2\bar{x}(t_1) + A_2\bar{G}_2 + B_2]\delta t_2\} \\
& - \bar{F}_3[A_3\bar{x}(t_2) + B_3]\delta t_2 \\
& + [\bar{F}_3 A_3\bar{x}(t_2) + A_3\bar{G}_3 + B_3]\delta t_3
\end{aligned}
$$

after rearranging the terms, one arrives at

$$\boxed{\begin{aligned} x(t_3) - \bar{x}(\bar{t}_3) =& \bar{F}_3\bar{F}_2\bar{F}_1[x(t_0) - \bar{x}(t_0)] \\ &+ \bar{F}_3\bar{F}_2[\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 - A_2\bar{x}(\bar{t}_1) + B_1 - B_2]\delta t_1 \\ &+ \bar{F}_3[\bar{F}_2 A_2\bar{x}(t_1) + A_2\bar{G}_2 - A_3\bar{x}(\bar{t}_2) + B_2 - B_3]\delta t_2 \\ &+ [\bar{F}_3 A_3\bar{x}(t_2) + A_3\bar{G}_3 + B_3]\delta t_3 \end{aligned}} \tag{4.24}$$

Now, there are enough iterations to deduce a pattern on each increment of the equation, so the equation for $x(t_i)$ can be deduced as

$$\boxed{\begin{aligned} x(t_i) - \bar{x}(\bar{t}_i) =& \bar{F}_i\bar{F}_{i-1}\ldots\bar{F}_1[x(t_0) - \bar{x}(t_0)] \\ &+ \bar{F}_i\bar{F}_{i-1}\ldots\bar{F}_1[\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 - A_2\bar{x}(\bar{t}_1) + B_1 - B_2]\delta t_1 \\ &+ \bar{F}_i\bar{F}_{i-1}\ldots\bar{F}_2[\bar{F}_2 A_2 x(t_1) + A_2\bar{G}_2 - A_3\bar{x}(\bar{t}_2) + B_2 - B_3]\delta t_2 \\ &\quad\vdots \\ &+ \bar{F}_i[\bar{F}_{i-1} A_{i-1}\bar{x}(t_{i-2}) + A_{i-1}\bar{G}_{i-1} - A_i\bar{x}(\bar{t}_{i-1}) + B_{i-1} - B_i]\delta t_{i-1} \\ &+ [\bar{F}_i A_i\bar{x}(t_i) + A_i\bar{G}_i + B_i]\delta t_i \end{aligned}} \tag{4.25}$$

and then, for $x(t_N)$ the equation is as follows

$$\begin{aligned} x(t_N) - \bar{x}(\bar{t}_N) =& \bar{F}_N\bar{F}_{N-1}\ldots\bar{F}_1[x(t_0) - \bar{x}(t_0)] \\ &+ \bar{F}_N\bar{F}_{N-1}\ldots\bar{F}_2[\overbrace{\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 - A_2\bar{x}(\bar{t}_1) + B_1 - B_2}^{\square}]\delta t_1 \\ &+ \bar{F}_N\bar{F}_{N-1}\ldots\bar{F}_3[\bar{F}_2 A_2 x(t_1) + A_2\bar{G}_2 - A_3\bar{x}(\bar{t}_2) + B_2 - B_3]\delta t_2 \\ &\quad\vdots \\ &+ \bar{F}_N[\bar{F}_{N-1} A_{N-1}\bar{x}(t_{N-2}) + A_{N-1}\bar{G}_{N-1} - A_N\bar{x}(\bar{t}_{N-1}) \\ &\qquad\qquad +B_{N-1} - B_N]\delta t_{N-1} \\ &+ [\underbrace{\bar{F}_N A_{N-1}\bar{x}(t_{N-1}) + A_N\bar{G}_N + B_N}_{\bigcirc}]\delta t_N \end{aligned} \tag{4.26}$$

The equation (4.26) can be developed further to remove the dependencies on the intermediary states $x(t_0), x(t_1), ..., x(t_{N-2})$. By expanding $\square$ and $\bigcirc$, the following identity holds

$$
\begin{aligned}
\square &= \bar{F}_1 A_1 x(t_0) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2 \\
&= A_1 \bar{F}_1 x(t_0) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2 \\
&= A_1 [\bar{x}(\bar{t}_1) - \bar{G}_1] + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2 \\
&= A_1 \bar{x}(\bar{t}_1) - \cancel{A_1 \bar{G}_1} + \cancel{A_1 \bar{G}_1} - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2 \\
&= (A_1 - A_2) \bar{x}(\bar{t}_1) + B_1 - B_2
\end{aligned} \tag{4.27}
$$

$$
\begin{aligned}
\bigcirc &= \bar{F}_N A_N \bar{x}(t_{N-1}) + A_N \bar{G}_N + B_N \\
&= A_N \bar{F}_N \bar{x}(t_{N-1}) + A_N \bar{G}_N + B_N \\
&= A_N [\bar{x}(\bar{t}_N) - \bar{G}_N] + A_N \bar{G}_N + B_N \\
&= A_N \bar{x}(\bar{t}_N) - \cancel{A_N \bar{G}_N} + \cancel{A_N \bar{G}_N} + B_N \\
&= A_N \bar{x}(\bar{t}_N) + B_N
\end{aligned} \tag{4.28}
$$

replacing (4.27) and (4.28) in (4.26)

$$
\boxed{
\begin{aligned}
x(t_N) - \bar{x}(\bar{t}_N) = \quad & \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_1 [x(t_0) - \bar{x}(t_0)] \\
& + \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_2 [(A_1 - A_2) \bar{x}(\bar{t}_1) + B_1 - B_2] \delta t_1 \\
& + \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_3 [(A_2 - A_3) \bar{x}(\bar{t}_2) + B_2 - B_3] \delta t_2 \\
& \quad \vdots \\
& + \bar{F}_N [(A_{N-1} - A_N) \bar{x}(\bar{t}_{N-1}) + B_{N-1} - B_N] \delta t_{N-1} \\
& + [A_N \bar{x}(\bar{t}_N) + B_N] \delta t_N
\end{aligned}
} \tag{4.29}
$$

and renaming the terms of (4.29) to

$$e(t) = x(t) - \bar{x}(\bar{t}) \tag{4.30}$$

$$\Phi = \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_1 \tag{4.31}$$

$$\begin{aligned} \Gamma = \Big[& \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_2 [(A_1 - A_2)\bar{x}(\bar{t}_1) + B_1 - B_2], \\ & \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_3 [(A_2 - A_3)\bar{x}(\bar{t}_2) + B_2 - B_3], \\ & \quad \vdots \\ & \bar{F}_N [(A_{N-1} - A_N)\bar{x}(\bar{t}_{N-1}) + B_{N-1} - B_N], \\ & A_N \bar{x}(\bar{t}_N) + B_N \Big] \end{aligned} \tag{4.32}$$

$$\delta t = [\delta t_1, \delta t_2, \dots, \delta t_N]^T \tag{4.33}$$

we get to the affine equation

$$\boxed{e(t_N) = \Phi e(t_0) + \Gamma \delta t} \tag{4.34}$$

that can be rewritten in terms of j cycle as

$$e(t_{j,N}) = \Phi e(t_{j,0}) + \Gamma \delta t[j] \tag{4.35}$$

where $\bar{x}$ is calculated as

$$\bar{x}(\bar{t}_1) = F_1 \bar{x}(\bar{t}_0) + G_1 \tag{4.36}$$

$$\bar{x}(\bar{t}_2) = F_2 \bar{x}(\bar{t}_1) + G_2 \tag{4.37}$$

$$\vdots$$

$$\bar{x}(\bar{t}_N) = F_N \bar{x}(\bar{t}_{N-1}) + G_N \tag{4.38}$$

and F_i and G_i are calculated as

$$F_i = e^{A_i(t_i - t_{i-1})} \tag{4.39}$$

$$G_i = \int_{\bar{t}_{i-1}}^{\bar{t}_i} e^{A_i(\bar{t}_i - \tau)} B_i d\tau \tag{4.40}$$

4.3 MPC Problem

Note that the equation (4.34) relates to the equation (3.1) as both describe the evolution of a system over time. The equation (3.1) represents the system's state and input at a given time step, and the equation (4.34) describes the error between the system's predicted and actual values at two different time points.

Referring to equation (3.1), we can make some substitutions to simplify our discussion. We'll replace variable x with e, and u with δt. Additionally, we'll denote the cycle index with k. With these substitutions in place, we arrive at equation (4.34).

4.3.1 Cost Function

The cost function for the MPC problem is given by

$$J = \left\| e[N_p] \right\|_{P_f}^2 + \sum_{j=0}^{N_p-1} \left\| e[j] \right\|_Q^2 + \left\| \delta t[j] \right\|_R^2 \tag{4.41}$$

where $Q = Q^T > 0$, $R = R^T > 0$, $e[j]$ is the error at cycle j and N_p is the prediction horizon spanning.

The error $e[j]$ is computed by equation (4.35) and the time vector $\delta t[j]$ is the difference between the switching times of the cycle and the nominal switching of the target trajectory

$$\delta t[j] = \left[t_{j,1} - \bar{t}_{j,1}, t_{j,2} - \bar{t}_{j,2}, ..., t_{j,N-1} - \bar{t}_{j,N-1} \right]^T \tag{4.42}$$

4.3.2 Dwell-time Constraints

We want to enforce a constraint on the dwell times where the switching times will the greater than a minimum value

$$(t_i - t_{i-1}) \geq \Delta t_{i,min} \ , \quad i = 1, 2, \dots, N \tag{4.43}$$

we can write this as

$$\delta t_1 \geq -\Delta \bar{t}_1 + \Delta t_{1,min} \tag{4.44}$$

$$\delta t_i - \delta t_{i-1} \geq -\Delta \bar{t}_i + \Delta t_{i,min} \ , \quad i = 1, 2, \dots, N \tag{4.45}$$

$$\delta t_{N-1} \geq -\Delta \bar{t}_N + \Delta t_{N,min} \tag{4.46}$$

with

$$\Delta\bar{t}_i = \bar{t}_i - \bar{t}_{i-1} \tag{4.47}$$

Rearranging this to a matrix form we have

$$L\delta t \geq c \tag{4.48}$$

with

$$L = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ 0 & 0 & 0 & \cdots & -1 & 1 \\ 0 & 0 & 0 & \cdots & 0 & -1 \end{bmatrix} \tag{4.49}$$

$$\delta t = \begin{bmatrix} \delta t_1 \\ \delta t_2 \\ \vdots \\ \delta t_{N-1} \\ \delta t_N \end{bmatrix} \tag{4.50}$$

$$c = \begin{bmatrix} -\Delta\bar{t}_1 + \Delta t_{1,min} \\ -\Delta\bar{t}_2 + \Delta t_{2,min} \\ \vdots \\ -\Delta\bar{t}_{N-1} + \Delta t_{N-1,min} \\ -\Delta\bar{t}_N + \Delta t_{N,min} \end{bmatrix} \tag{4.51}$$

This constraint is related to the jth cycle of the system, so the notation of δt can be rewritten as

$$\delta t[j] = \begin{bmatrix} \delta t_{j,1} \\ \delta t_{j,2} \\ \vdots \\ \delta t_{j,N-1} \\ \delta t_{j,N} \end{bmatrix} \tag{4.52}$$

To use this with the constraints $S\hat{\mathbf{u}}[k] \leq b$, this can be rearranged as

$$-L\delta t[j] \leq -c$$

4.4 Application Examples

4.4.1 Double Integrator

4.4.1.1 Parameters of the Simulation

This example describes the double integrator model used in Marcolino (2018) of the form

$$\dot{x}(t) = A_x x(t) + B_c u(t) \tag{4.53}$$

where

$$A_c = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, B_c = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{4.54}$$

We are using the associated input signal computed by Marcolino *et al.* (2021) and the resulting trajectory is defined by

$$\bar{u} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, T_s = \begin{bmatrix} 0 & 2 & 3 & 5 & 6 \end{bmatrix} \tag{4.55}$$

where $\bar{u}$ is the control command and T_s is the switching time values for the cyclic trajectory, illustrated in Figure 4.2.

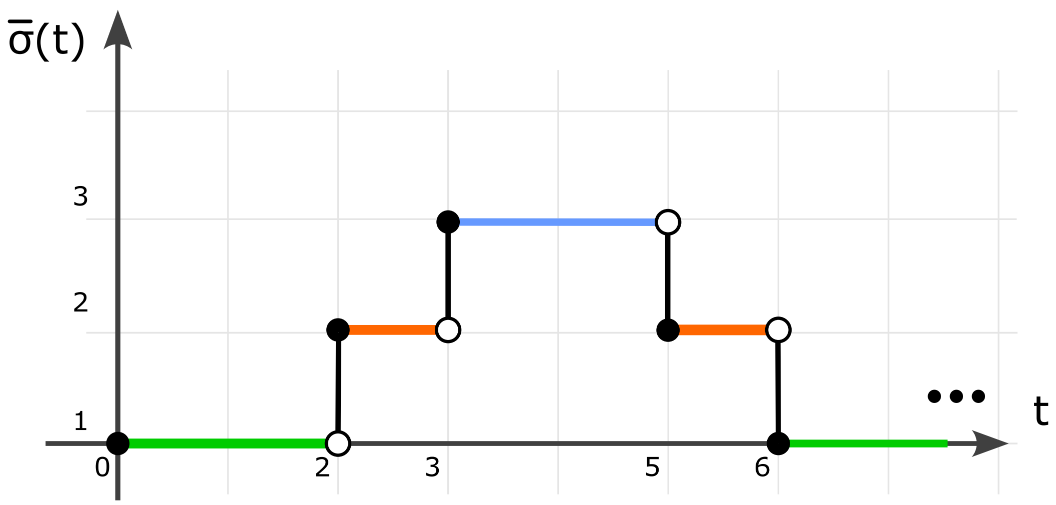

FIGURE 4.2 – Reference signal for Double Integrator

The original problem has the same dynamics during all operations, so we need to construct the associated A_i, b_i for each operation mode. One solution is to create 3 unique pairs

$$A_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad A_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad A_3 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad b_3 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

and the index for the operation mode $\Omega = \begin{bmatrix} 1, 2, 3, 2 \end{bmatrix}$, with reference state vector for this trajectory is $\bar{x}_{ref} = \begin{bmatrix} 2, -1 \end{bmatrix}$, the maximum value for the period of the trajectory cycle $T_{p_{max}} = 1$ and initial state of the nominal target trajectory $x_0 = \begin{bmatrix} 1.5, -1 \end{bmatrix}$.

4.4.1.2 Prediction Matrices

From (4.32) and (4.33) we construct

$$\Phi = \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix} \tag{4.56}$$

$$\Gamma = \begin{bmatrix} 4 & 3 & -1 \\ 1 & 1 & -1 \end{bmatrix} \tag{4.57}$$

with the switching time constraints

$$c = \begin{bmatrix} -\Delta t_{r_1} + \Delta t_{r_{1,min}} \\ -\Delta t_{r_2} + \Delta t_{r_{2,min}} \\ \vdots \\ -\Delta t_{r_{N-1}} + \Delta t_{r_{N-1,min}} \\ -\Delta t_{r_N} + \Delta t_{r_{N,min}} \end{bmatrix} = \begin{bmatrix} -2 + 1.625 \\ -1 + 0.635 \\ -2 + 1.625 \\ -1 + 0.625 \end{bmatrix} = \begin{bmatrix} -0.375 \\ -0.365 \\ -0.375 \\ -0.365 \end{bmatrix} \tag{4.58}$$

4.4.1.3 Results

Figure 4.3 presents the feasibility region $\mathcal{D}$ for three different prediction horizons ($N_p = 1$, $N_p = 2$ and $N_p = 3$) as defined in Section 3.5. A larger prediction horizon reflected in a larger feasibility region enables the use of the predictive control law with initial conditions farther from the target.

The simulation uses the system (4.54) with the command signal (4.55) and with a

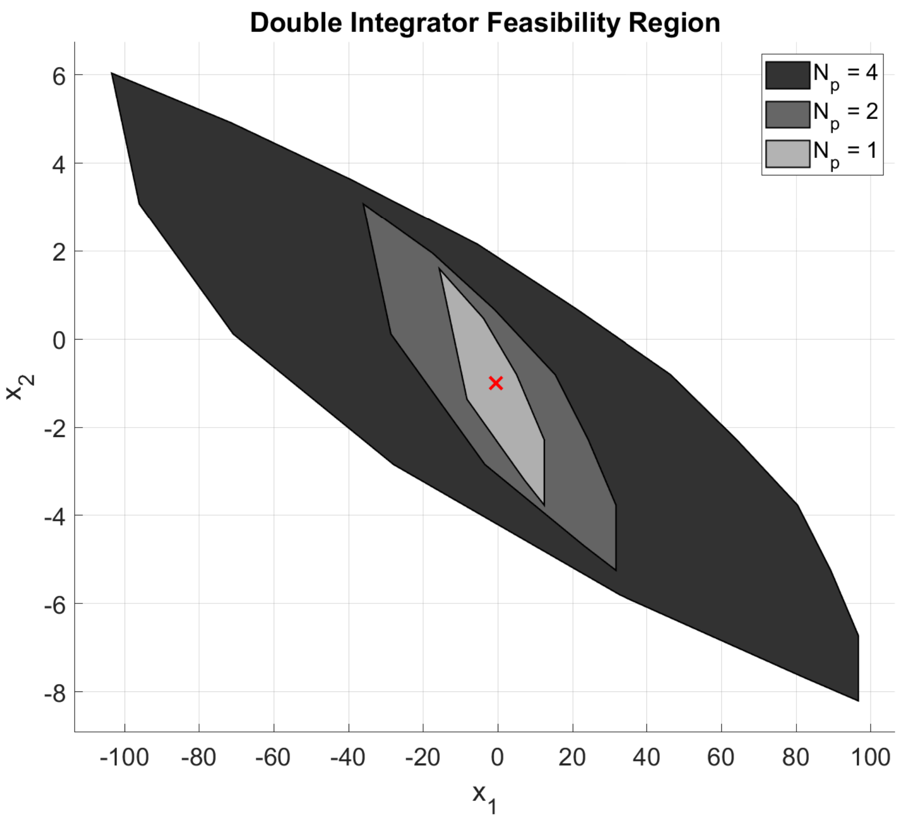

FIGURE 4.3 – Feasibility Region Double Integrator

initial condition $x_0 = \begin{bmatrix} 1.5\ V, -0.5\ A \end{bmatrix}$. Figure 4.4 shows the simulation without the MPC control being applied running for 5 cycles and we can see that the the trajectory is diverging from the initial position of the target cyclic trajectory $\bar{x}_0 = \begin{bmatrix} -0.5\ V, -1.0\ A \end{bmatrix}$ position.

Now, for the same condition, we can turn on the MPC controller and run the simulation for 50 cycles. Figure 4.5 shows the resulting trajectory and Figure 4.6 shows the associated command signal. We can see that the controller drives the trajectory of the system to the target cyclic trajectory, we also that the command signal gets in sync with the reference command.

4.4.2 Buck-boost converter

4.4.2.1 Parameters of the Simulation

This example describes the Buck-Boost converter model described in session 2.3.1, the associated mode sequence for the target cyclic trajectory is shown in Figure 4.7

where

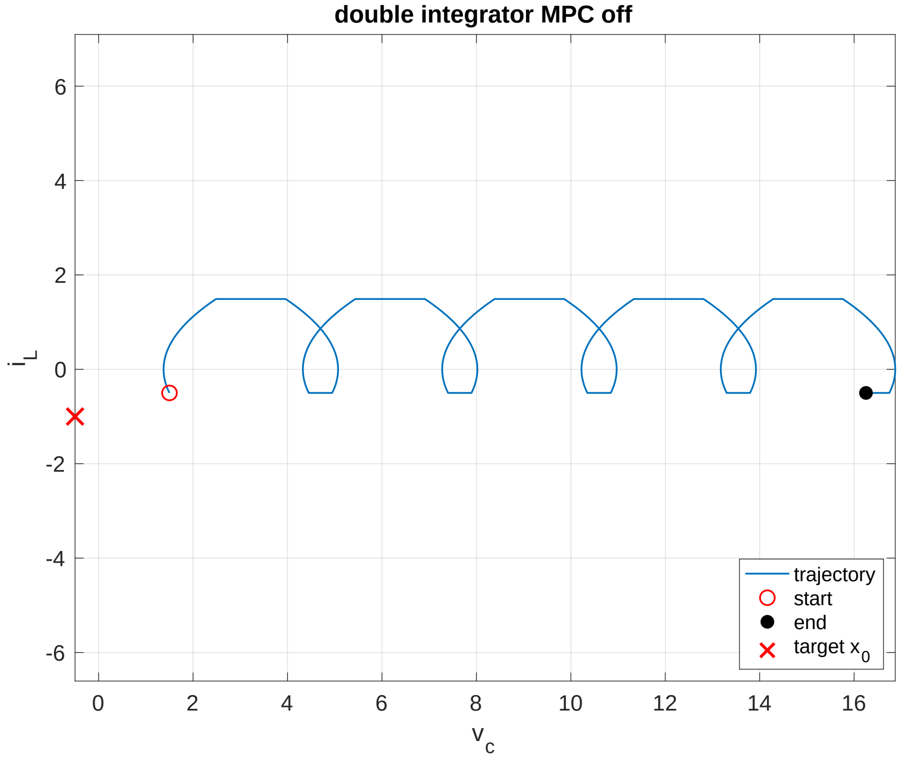

FIGURE 4.4 – State Trajectory MPC off, 4 cycles

$$A_1 = \begin{bmatrix} 0 & 0 \\ 0 & -1/(RC) \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & 1/L \\ -1/C & -1/(RC) \end{bmatrix}$$

$$b_1 = \begin{bmatrix} E/L \\ 0 \end{bmatrix} \qquad b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and the operation mode is characterized by the index $\Omega = [1, 2]$. The time switch values are defined as $\tau^\infty = [0, 0.2513, 0.5914]$ s, and the reference state vector for this trajectory is denoted as $\bar{x}_{ref} = [2\text{ V}, -1\text{ A}]$. Additionally, the maximum period value for the trajectory cycle is specified as $T_{p_{max}} = 1$ s, while the initial state of the nominal target trajectory is given by $x_0 = [1.8708\text{ V}, -1.1198\text{ A}]$. The time step of the simulation if $t_{step} = 1 \times 10^{-5}$ s and the weight matrix used in the computation of the trajectory is Q is a diagonal matrix of ones.

4.4.2.2 Prediction Matrices

The prediction matrices are constructed using (4.32) and (4.33), and the result is

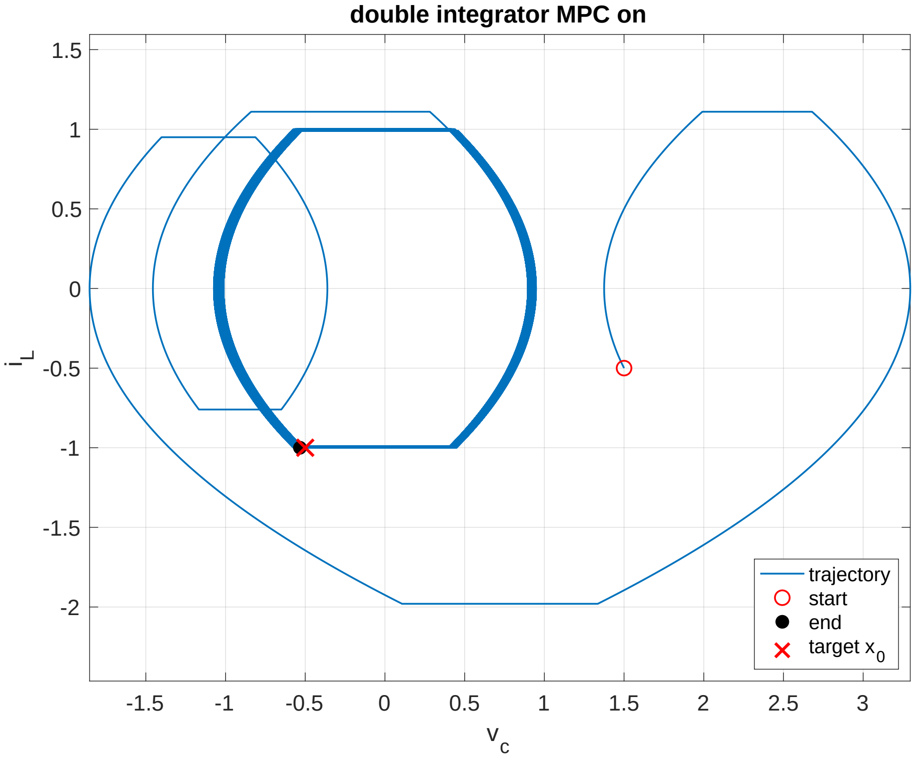

FIGURE 4.5 – State Trajectory MPC on, 50 cicles

$$\Phi = \begin{bmatrix} 0.9713 & 0.2192 \\ -0.1707 & 0.5857 \end{bmatrix} \tag{4.59}$$

$$\Gamma = \begin{bmatrix} 2.4556 \\ 0.9280 \end{bmatrix} \tag{4.60}$$

with the switching time constraints

$$c = \begin{bmatrix} -\Delta t_{r_1} + \Delta t_{r_{1,min}} \\ -\Delta t_{r_2} + \Delta t_{r_{2,min}} \\ \vdots \\ -\Delta t_{r_{N-1}} + \Delta t_{r_{N-1,min}} \\ -\Delta t_{r_N} + \Delta t_{r_{N,min}} \end{bmatrix} = \begin{bmatrix} -0.25 + 0.25 \\ -0.2504 + 0.25 \end{bmatrix} = \begin{bmatrix} 0 \\ -0.04 \end{bmatrix} \tag{4.61}$$

4.4.2.3 Results

Figure 4.8 presents the feasibility region $\mathcal{D}$ for three different prediction horizons ($N_p = 1$, $N_p = 2$ and $N_p = 3$) for the Buck-Boost Converter.

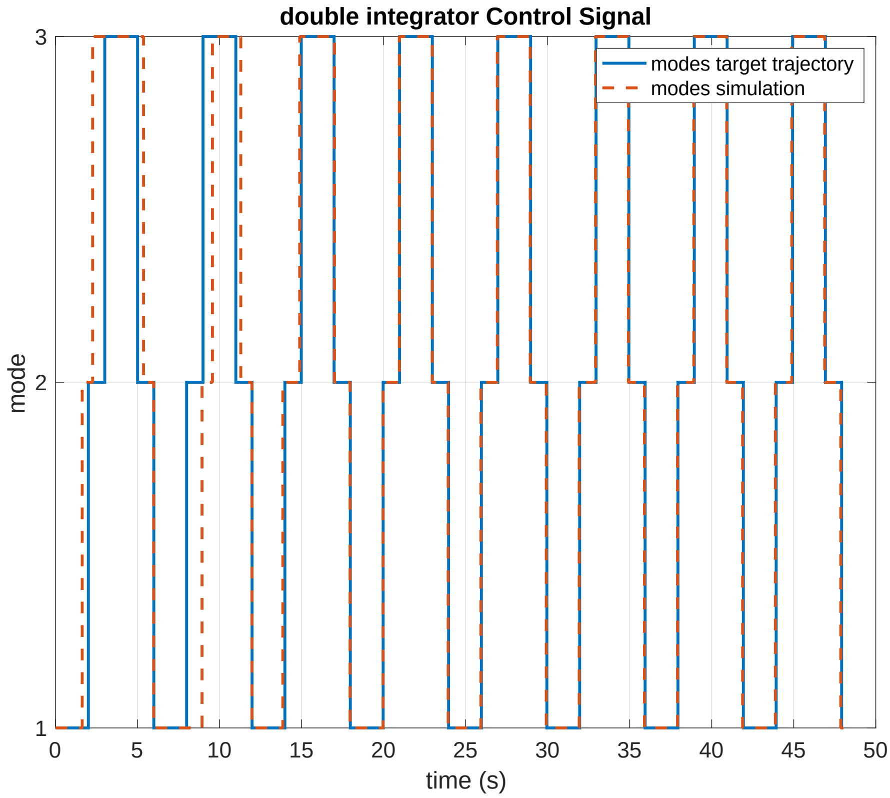

FIGURE 4.6 – Operation mode switch

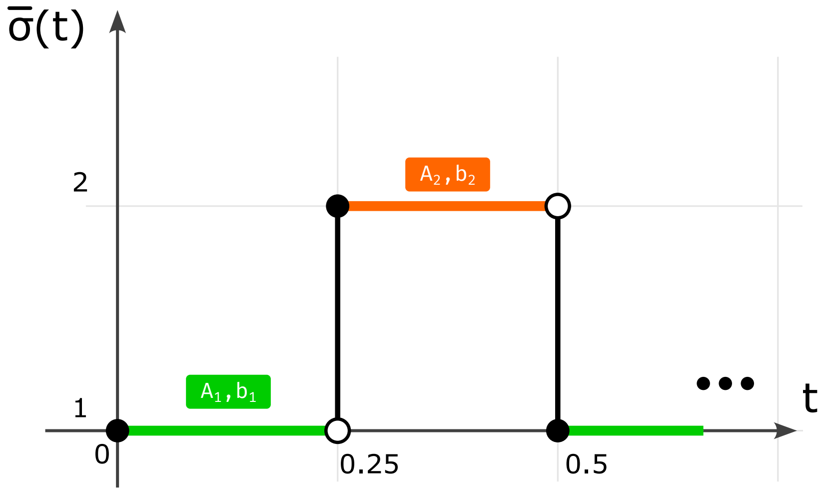

FIGURE 4.7 – Reference sequence for Buck-Boost converter

Figure 4.9 shows the simulation of the trajectory for the system , in blue we see the trajectory without using the MPC controller, and in orange we see the trajectory using the MPC controller. They are both starting from $x_0 = \left[2.0493 \text{ V}, -0.6108 \text{ A}\right]$ and converge to the initial position of the target cyclic trajectory $\bar{x}_0 = \left[1.8493 \text{ V}, -1.1108 \text{ A}\right]$. Note that the trajectory with the MPC controller converges faster to the cyclic trajectory.

Figure 4.10 shows the control signal sequence, where we can see that the difference

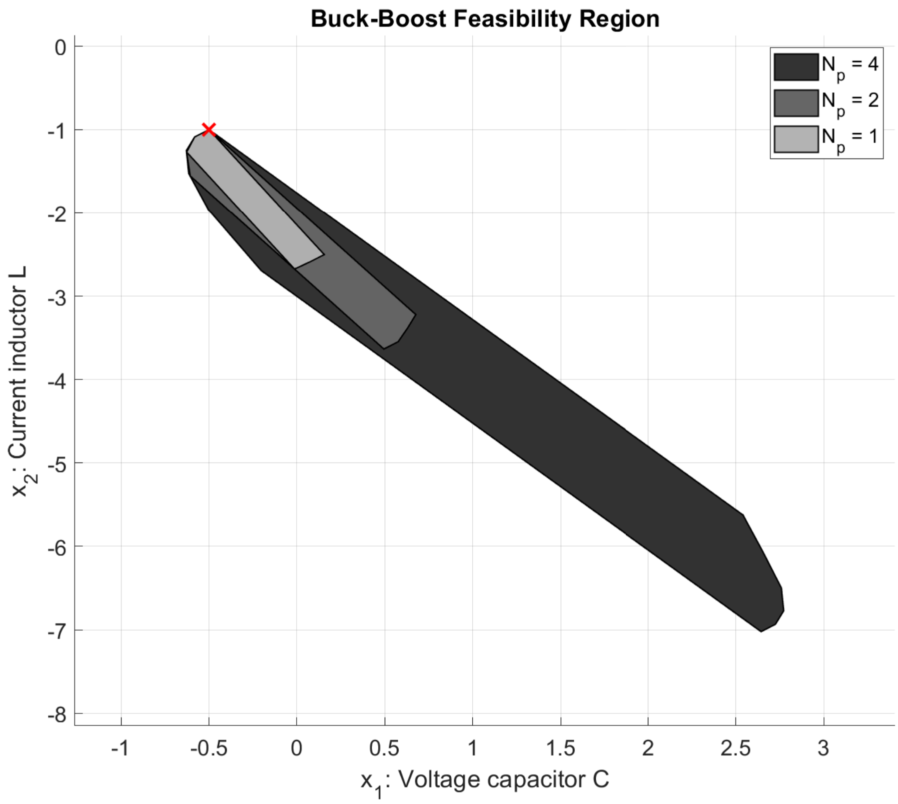

FIGURE 4.8 – Buck-Boost Feasibility Region antes

between the nominal target trajectory and the controlled trajectory decrease as they get in sync with each other.

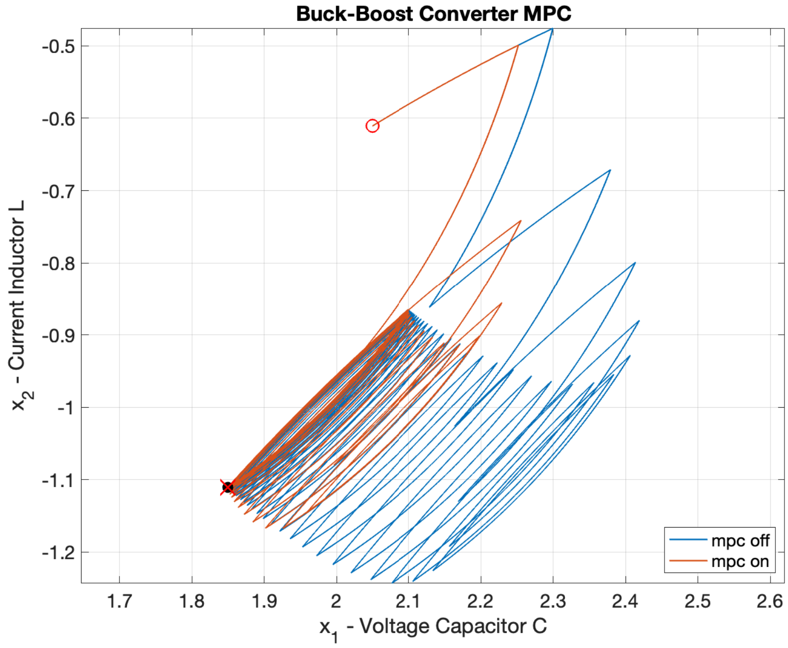

FIGURE 4.9 – State Trajectory for Buck-Boost converter

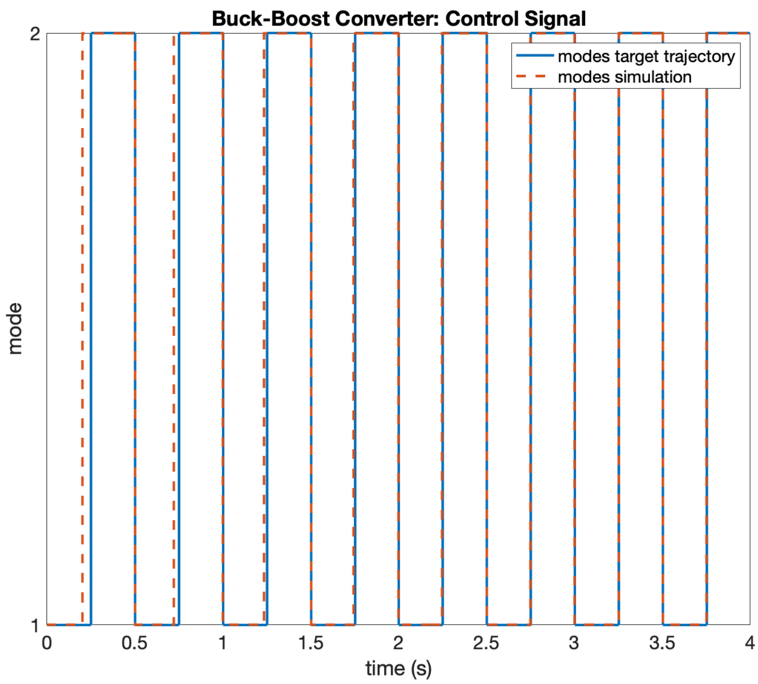

FIGURE 4.10 – Control Signal Sequence for Buck-Boost converter

4.4.3 Multilevel converter

4.4.3.1 Parameters of the Simulation

This example describes the Multilevel converter model described in session 2.3.2, the dynamics of this system are defined by

$$A_i = \begin{bmatrix} 0 & 0 & \Big(u_2(i) - u_1(i)\Big)/C1 \\ 0 & 0 & \Big(u_3(i) - u_2(i)\Big)/C2 \\ \Big(u_1(i) - u_2(i)\Big)/L & \Big(u_2(i) - u_3(i)\Big)/L & -R/L \end{bmatrix}$$
$$b_i = \begin{bmatrix} 0 \\ 0 \\ \Big(E/L\Big)u_3(i) \end{bmatrix} \tag{4.62}$$

with $R = 10.0\ \Omega$, $L = 10.0\ \mathrm{mH}$, $C_1 = 40.0\ \mu\mathrm{F}$, $C_2 = 40.0\ \mu\mathrm{F}$, $E = 30.0\ \mathrm{V}$

and the associated mode sequence for the target cyclic trajectory is shown in Figure 4.11.

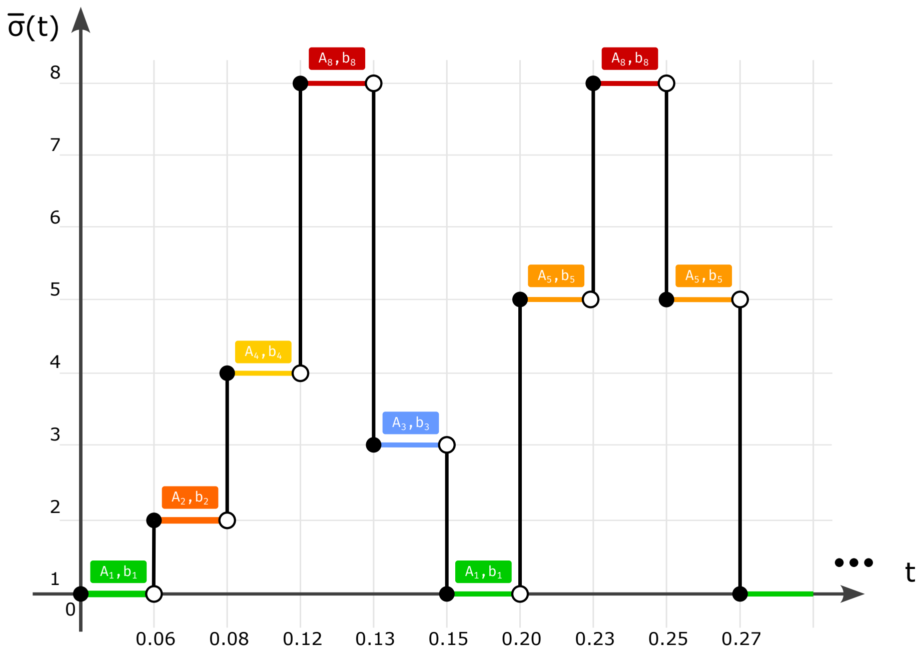

FIGURE 4.11 – Reference sequence for Multilevel converter

with $u(i)$ being

TABLE 4.1 – Values of u_i for the Multilevel converter

mode (i)	u_1	u_2	u_3
1	0	0	0
2	0	0	1
3	0	1	0
4	0	1	1
5	1	0	0
6	1	0	1
7	1	1	0
8	1	1	1

and the operation mode is characterized by the index $\Omega = \left[1, 2, 4, 8, 3, 1, 5, 8, 5\right]$. The time switch values are defined as $\tau^{\infty} = \left[0, 0.0639, 0.0878, 0.1119, 0.1344, 0.1567, 0.2065, 0.23, 0.2529, 0.2756\right]$ ms, and the reference state vector for this trajectory is denoted as $\bar{x}_{ref} = \left[10\text{ V}, 20\text{ V}, 1\text{ A}\right]$. Additionally, the maximum period value for the trajectory cycle is specified as $T_{p_{max}} = 4 \times 10^{-4}$, while the initial state of the nominal target trajectory is given by $x_0 = \left[9.5143\text{ V}, 19.8211\text{ V}, 1.0314\text{ A}\right]$. The time step of the simulation if $t_{step} = 1 \times 10^{-7}$ s and the weight matrix used in the computation of the trajectory is $Q = diag([10, 5, 20000])$.

4.4.3.2 Prediction Matrices

The prediction matrices are constructed using (4.32) and (4.33), and the result is

$$\Phi = \begin{bmatrix} 0.999539 & 0.001900 & -0.124610 \\ -0.002031 & 0.999915 & 0.000764 \\ -0.000461 & -0.000290 & 0.759578 \end{bmatrix} \tag{4.63}$$

$$\Gamma = \begin{bmatrix} 0.007784 & -2.578391 & -0.081618 \\ -2.400410 & 2.472097 & -0.079094 \\ 2.477894 & 0.042370 & -0.076050 \\ -2.602774 & 2.388977 & 0.171509 \\ 2.472432 & -2.594318 & 0.102211 \\ 2.703328 & 0 & -0.087828 \\ -2.354659 & 0 & -0.189865 \\ 2.354483 & 0 & 0.199932 \end{bmatrix}^T \times 10^4 \tag{4.64}$$

with the switching time constraints

$$c = \begin{bmatrix} -\Delta t_{r_1} + \Delta t_{r_{1,min}} \\ -\Delta t_{r_2} + \Delta t_{r_{2,min}} \\ \vdots \\ -\Delta t_{r_{N-1}} + \Delta t_{r_{N-1,min}} \\ -\Delta t_{r_N} + \Delta t_{r_{N,min}} \end{bmatrix} = \begin{bmatrix} -0.638811 + 0.20 \\ -0.239318 + 0.20 \\ -0.240440 + 0.20 \\ -0.225094 + 0.20 \\ -0.223339 + 0.20 \\ -0.497696 + 0.20 \\ -0.235714 + 0.20 \\ -0.228086 + 0.20 \\ -0.227575 + 0.20 \end{bmatrix} \times 10^{-4} = \begin{bmatrix} -0.438811 \\ -0.039318 \\ -0.040440 \\ -0.025094 \\ -0.023339 \\ -0.297696 \\ -0.035714 \\ -0.028086 \\ -0.027575 \end{bmatrix} \times 10^{-4} \tag{4.65}$$

4.4.3.3 Results

Figure 4.12 presents the feasibility region $\mathcal{D}$ for three different prediction horizons ($N_p = 1$, $N_p = 2$ and $N_p = 4$) for the Multilevel Converter.

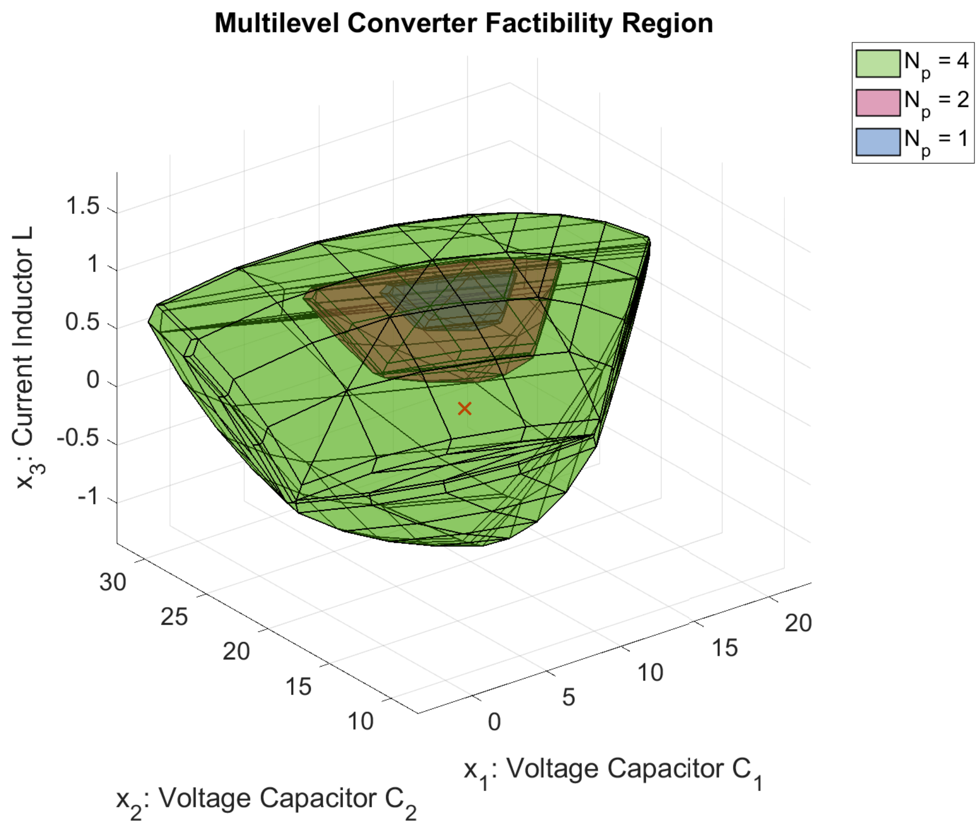

FIGURE 4.12 – Multilevel Converter: Feasibility Region

Figure 4.13 shows the trajectory for the Multilevel converter, in blue we see the trajectory without using the MPC controller, and in orange we see the trajectory using the MPC controller. They are starting from $x_0 = \left[7.5143\ \mathrm{V}, 20.8211\ \mathrm{V}, 0.0314\ \mathrm{A}\right]$, the trajectory with the MPC off goes to a point far from the targeted position $\bar{x}_0 = \left[9.5143\ \mathrm{V}, 19.8211\ \mathrm{V}, 1.0314\ \mathrm{A}\right]$, but the trajectory with the MPC on goes drives the system to the desired target.

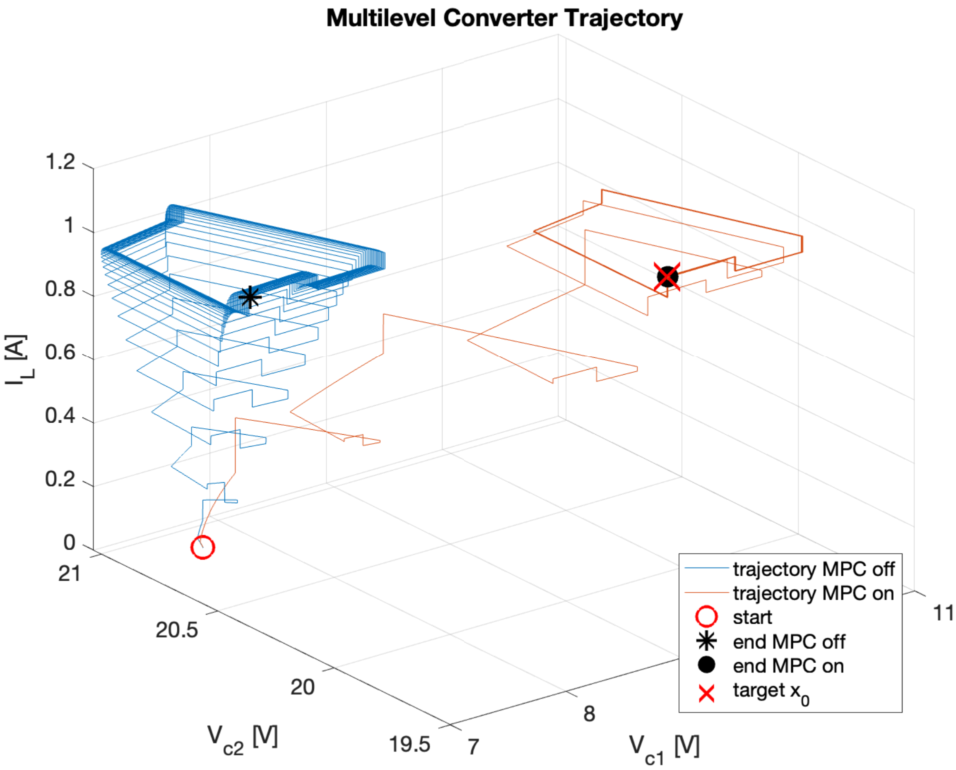

FIGURE 4.13 – Multilevel Converter: State Trajectory, $x_0 = \left[7.51, 20.82, 0.03\right]^T$

Note that the trajectory of Figure 4.13 does not converge to the same periodic trajectory. In Figure 4.14 we run the simulation with a different initial condition x_0 and we see that the trajectory with the MPC turned off converges to yet another periodic trajectory.

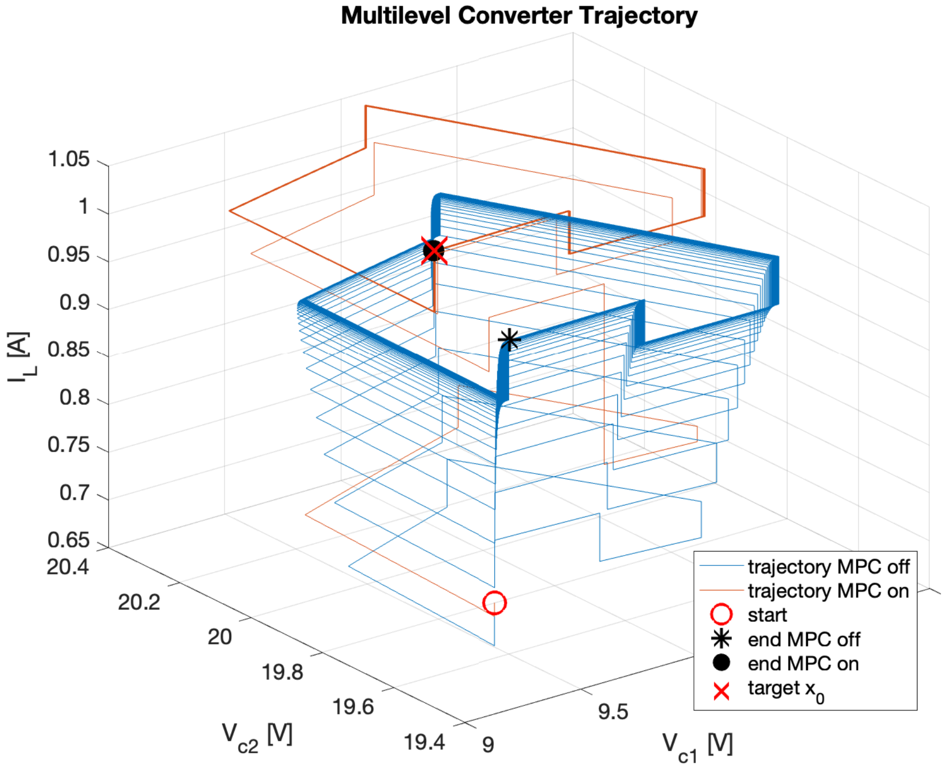

FIGURE 4.14 – Multilevel Converter: State Trajectory, $x_0 = \left[9.31, 19.52, 0.73\right]^T$

Figure 4.15 displays the states of the system over time.

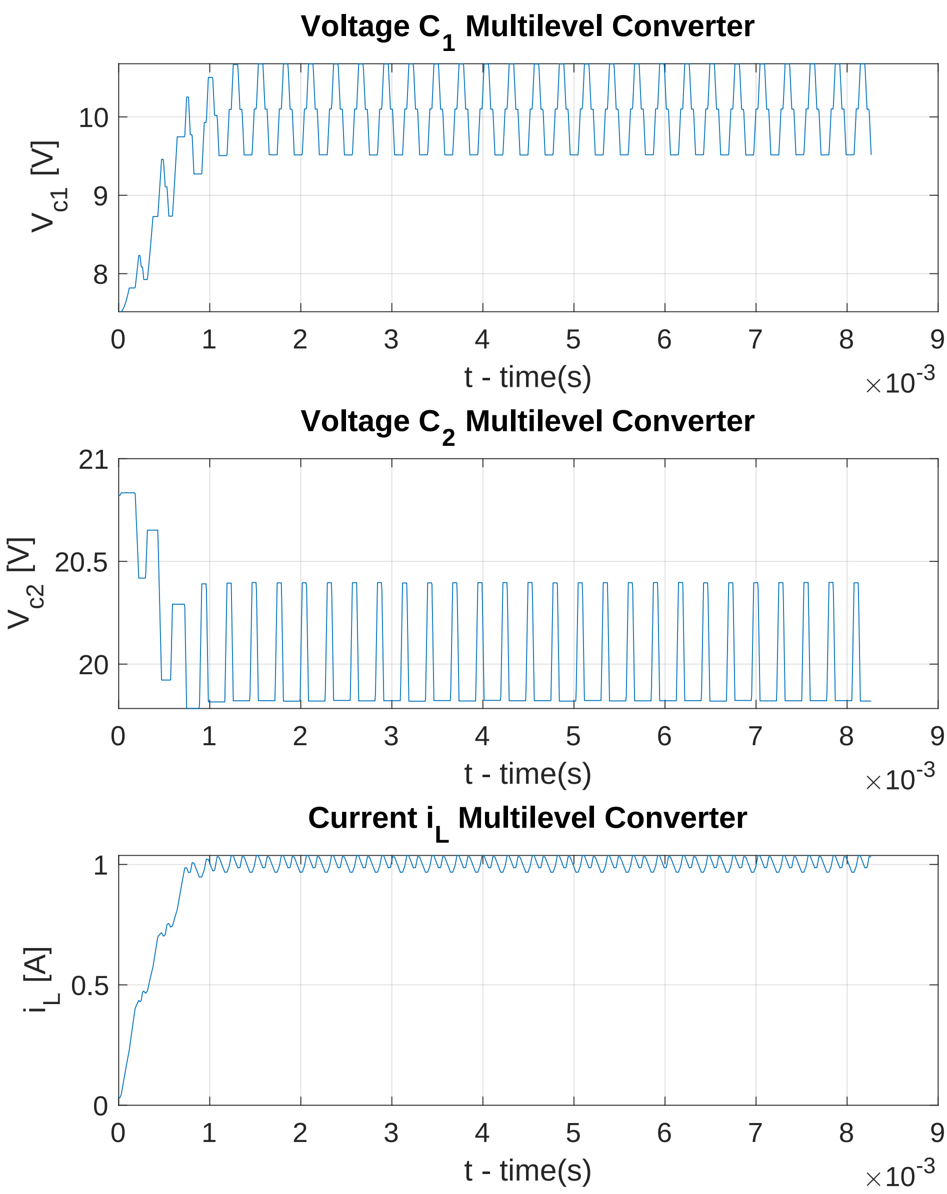

FIGURE 4.15 – Multilevel Converter: States over time

Figure 4.16 displays the control signal sequence, we can see the difference between the open loop and the controlled signal decreasing and getting in sync.

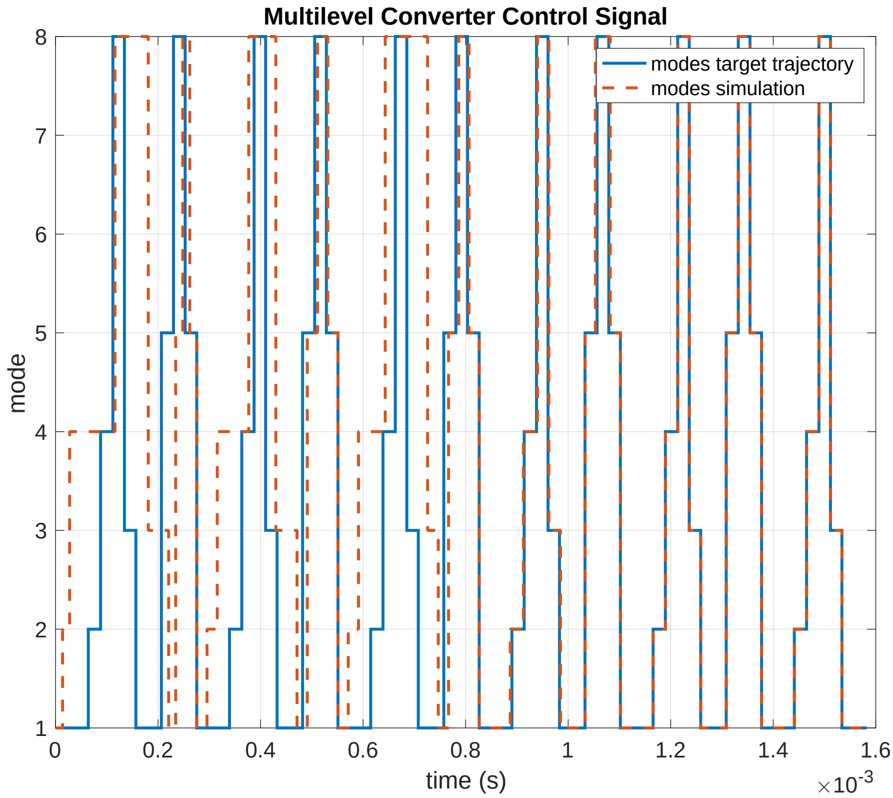

FIGURE 4.16 – Multilevel Converter: Control Signal

5 Conclusion

This work consists of extending the solution of controlling a switched affine system subjected to dwell-time constraints to control this class of systems also with changing dynamics at switching times. This allows the method to be used in more control problems and maintains the convenience of solving the control problem with quadratic programming convex optimization, which is a mature class of optimization problems.

To test the method, first, a double integrator system with fixed dynamics was implemented in session 4.4.1 to validate the new solution. The result shows that the system diverges when not in the cyclic trajectory without the MPC controller and the controller is capable of correcting the trajectory and eventually engaging with the target trajectory.

The application example in session 4.4.2 introduces a system with changing dynamics at each switching time. A cyclic trajectory was calculated in session 2.3.1 to use as an input parameter to implement the control method. In this example, the circuit has stable dynamics that converge to the cyclic trajectory. Using the MPC controller reduces the time to converge as can be seen in the comparison of the trajectories.

The multilevel converter in session 4.4.3 is a circuit that introduces more complexity, there are more operation modes and even modes that are repeated during the same cycle. The simulation without the MPC shows a trajectory that converges to a cyclic limit that is different than the target cyclic trajectory. With the use of the MPC controller, we see that the system is driven to the target trajectory. The limit cycle of this circuit is also changing with different initial conditions, in a first analysis this could be due to a numerical error in the simulation given that the numbers for this example are very small and susceptible to float point precision errors, but more work is needed to explain this behavior.

Future work will include the reimplementation of the linearization using augmented states, this approach was used by Patino *et al.* (2010) to write the system in an autonomous form and some initial analysis shows that this approach can simplify the linearization procedure to obtain the prediction equation for the cycle error and would contribute with improvement to the current work.

Another important work to do on top of the current state is to implement a physical in-lab multilevel circuit and apply the control solution in an embedded environment, the majority of the references in the topic treated the problem in the simulation domain and a real application would enrich the work. Also on this topic of thinking in real applications, future work contemplates the implementation in the simulation of the clock cycles, this means that the instants to switch the operation mode would be multiplies of the computer time, in circuits with time constants that are very small like the multilevel converter example this would cause a rounding error that would lead to perturbation being injected into the circuit.

We propose the time to finish the project to be in 3 semesters. The first one is to finish the theoretical dependencies and analysis of the digital domain of the work, the second is to implement the real circuit in-lab and the last is to wrap all results in the final work.

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TD	25 de janeiro de 2024	DCTA/ITA/DM-018/2015	68

5. TÍTULO E SUBTÍTULO:

Predictive control of switched affine systems subjected to dwell time constraints

6. AUTOR(ES):

Daniel Vieira

7. INSTITUIÇÃO(ÕES)/ÓRGÃO(S) INTERNO(S)/DIVISÃO(ÕES):

Instituto Tecnológico de Aeronáutica – ITA

8. PALAVRAS-CHAVE SUGERIDAS PELO AUTOR:

MPC; Model Predictive Control; Switched Affine System; Dwell-time Constraint

9. PALAVRAS-CHAVE RESULTANTES DE INDEXAÇÃO:

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10. APRESENTAÇÃO: (**X**) **Nacional** ( ) **Internacional**

ITA, São José dos Campos. Curso de Doutorado. Programa de Pós-Graduação em Engenharia Eletrônica e Computação. Área de Sistemas Aeroespaciais e Mecatrônica. Orientador: Prof. Dr. Roberto Kawakami Harrop Galvão. Doralice Serra. Defesa em 25/01/2024. Publicada em 2024.

11. RESUMO:

Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o sistema à trajetória cíclica alvo.

12. GRAU DE SIGILO:

(**X**) **OSTENSIVO** ( ) **RESERVADO** ( ) **SECRETO**